



SRH University

**Modulhandbuch
Studiengang
Applied Data Science and
Artificial Intelligence
M.Sc.**

Curriculum Master of Science (M.Sc.) Applied Data Science and Artificial Intelligence

(Version: Master of Science (M.Sc.) Applied Data Science and Artificial Intelligence | English | DA-ID:)

Module	Prüfung	Semester			
		1	2	3	4
Semester 1: Mandatory Modules Heidelberg, Hamburg, München, Fuerth					
First Steps into Data Science (6 ECTS)	Projektarbeit (60%) Lerntagebuch (40%)	6			
Python Foundations for Data Science (6 ECTS)	Praktische Arbeit (100%)	6			
Data Engineering (6 ECTS)	Projektarbeit (100%)	6			
Statistics and Machine Learning (6 ECTS)	Projektarbeit (100%)	6			
Semester 1: General or Interdisciplinary Track Campus Specific					
General Track Heidelberg					
Advanced Python for Data Science (6 ECTS)	Praktische Arbeit (100%)	6			
Creative AI and Media Analytics Heidelberg					
Recommendation Systems and Media Personalization (6 ECTS)	Projektarbeit (100%)	6			
Supply Chain and Logistics Analytics Hamburg					
Demand Forecasting (6 ECTS)	Projektarbeit (100%)	6			
Business Analytics Fürth					
Trend Analysis and Consumer Behaviour (6 ECTS)	Studienarbeit (100%)	6			
AI-Driven Bioinformatics & Life Sciences München					
Precision Medicine (6 ECTS)	Projektarbeit (100%)	6			
Semester 2: Mandatory Modules Heidelberg, Hamburg, München, Fuerth					
Data Management (6 ECTS)	Projektarbeit (100%)		6		
Natural Language Processing (6 ECTS)	Praktische Arbeit (100%)		6		
Responsible AI and Data Ethics (6 ECTS)	Projektarbeit (70%) Präsentation (30%)		6		
Semester 2: General or Interdisciplinary Track Campus Specific					
General Track Heidelberg					
Collaborative Industry Project - I (6 ECTS)	Projektarbeit (100%)		6		
Collaborative Industry Project - II (6 ECTS)	Projektarbeit (100%)		6		
Creative AI and Media Analytics Heidelberg					
Social Media Analytics and Sentiment Analysis (6 ECTS)	Projektarbeit (100%)		6		
Generative AI for Creative Content (6 ECTS)	Projektarbeit (70%) Präsentation (30%)		6		
Supply Chain and Logistics Analytics Hamburg					
Inventory Analytics (6 ECTS)	Projektarbeit (70%) Präsentation (30%)		6		

Logistics and Transportation Analytics (6 ECTS)	Projektarbeit (100%)		6		
Business Analytics Fuerth					
Retail and E-commerce Analytics (6 ECTS)	Präsentation (100%)		6		
Performance and Engagement Analytics (6 ECTS)	Studienarbeit (100%)		6		
AI-Driven Bioinformatics & Life Sciences Munich					
AI for Drug Discovery (6 ECTS)	Projektarbeit (100%)		6		
Biomedical image analysis (6 ECTS)	Praktische Arbeit (100%)		6		
Semester 3: Elective Semester 3. Semester/Internship/Semester Abroad					
Internship/Semester Abroad (30 ECTS)	Präsentation (30%) Praxisbericht (50%) Lerntagebuch (20%)			30	
Deep Learning (6 ECTS)	Praktische Arbeit (100%)			6	
Cloud Computing and Data Architectures (6 ECTS)	Projektarbeit (100%)			6	
Data Visualization and Storytelling (6 ECTS)	Projektarbeit (100%)			6	
Entrepreneurship (6 ECTS)	Businessplan (70%) Präsentation (30%)			6	
Semester 3: General or Interdisciplinary Track Campus Specific					
General Track Heidelberg					
AI Vision (6 ECTS)	Praktische Arbeit (100%)			6	
Creative AI and Media Analytics Heidelberg					
Gaming Analytics (6 ECTS)	Projektarbeit (100%)			6	
Supply Chain and Logistics Analytics Hamburg					
Risk and Sustainability Analytics (6 ECTS)	Projektarbeit (70%) Präsentation (30%)			6	
Business Analytics Fuerth					
Strategic Communication and Leadership (6 ECTS)	Präsentation (100%)			6	
AI-Driven Bioinformatics & Life Sciences (Munich) Munich					
Biomedical image analysis (6 ECTS)	Praktische Arbeit (100%)			6	
Semester 4: Mandatory Modules Heidelberg, Hamburg, München, Fuerth					
Master Thesis Seminar (6 ECTS)	Exposé (100%)				6
Master Thesis (24 ECTS)	Thesis (75%) Kolloquium (25%)				24
ECTS insgeamt	120	30	30	30	30

Inhalt

Semester 1: Mandatory Modules	1
First Steps into Data Science	2
Python Foundations for Data Science	6
Data Engineering	11
Statistics and Machine Learning	17
Semester 1: General or Interdisciplinary Track	21
General Track	22
Advanced Python for Data Science	23
Creative AI and Media Analytics	26
Recommendation Systems and Media Personalization	27
Supply Chain and Logistics Analytics	32
Demand Forecasting	33
Business Analytics	38
Trend Analysis and Consumer Behaviour	39
AI-Driven Bioinformatics & Life Sciences	42
Precision Medicine	43
Semester 2: Mandatory Modules	46
Data Management	47
Natural Language Processing	51
Responsible AI and Data Ethics	56
Semester 2: General or Interdisciplinary Track	61
General Track	62
Collaborative Industry Project - I	63
Collaborative Industry Project - II	67
Creative AI and Media Analytics	70
Social Media Analytics and Sentiment Analysis	71
Generative AI for Creative Content	74
Supply Chain and Logistics Analytics	78
Inventory Analytics	79
Logistics and Transportation Analytics	83
Business Analytics	86
Retail and E-commerce Analytics	87
Performance and Engagement Analytics	90
AI-Driven Bioinformatics & Life Sciences	93
AI for Drug Discovery	94
Biomedical image analysis	97
Semester 3: Elective Semester	101
Internship/Semester Abroad	102
Deep Learning	105
Cloud Computing and Data Architectures	110
Data Visualization and Storytelling	115
Entrepreneurship	119
Semester 3: General or Interdisciplinary Track	123
General Track	124
AI Vision	125
Creative AI and Media Analytics	130
Gaming Analytics	131

Supply Chain and Logistics Analytics	135
Risk and Sustainability Analytics	136
Business Analytics	139
Strategic Communication and Leadership	140
AI-Driven Bioinformatics & Life Sciences (Munich)	143
Biomedical image analysis	144
Semester 4: Mandatory Modules	148
Master Thesis Seminar	149
Master Thesis	152



Semester 1: Mandatory Modules

Heidelberg, Hamburg, München, Fuerth

Modulbeschreibung

First Steps into Data Science

Modulnummer	Niveau Master	Studiengangsleitung Prof. Dr. Swati Chandna
ECTS (Benotung) 6 (benotet)	Dauer Semester	Häufigkeit 2 per Year
Sprache English	Workload Insgesamt: 150 Stunden - davon Kontaktzeit: 48 Stunden - davon Selbststudium: 102 Stunden - davon Praxis: 0 Stunden	Veranstaltungstyp Fallarbeit Interaktives Seminar Selbststudium Teamarbeit

Teilnahmevoraussetzungen

Kursinformation

Modulprüfung

Prüfungsteile	Gewichtung (Bewertung)
Projektarbeit	60%
Lerntagebuch	40%

Qualifikationsziele (Learning Outcomes) und Kompetenzen

Nach erfolgreichem Abschluss des Moduls können die Studierenden ...

Handlungskompetenz

This module provides an applied, hands-on introduction to the data science mindset and process. Students will explore real-world data problems through structured frameworks (CRISP-DM, KDD), basic data visualization, and collaborative data storytelling. It emphasizes team-based exploration, ethical awareness, and reflective learning in navigating the ambiguity of real data

Fachkompetenz „Wissen und Verstehen“

- Describe the steps of data science pipelines: CRISP-DM, KDD, and their relevance
- Understand core stages: business understanding, data understanding, preparation, modeling, evaluation, deployment
- Conduct exploratory data analysis (EDA) on real datasets
- Explain basic data visualization and storytelling principles

Methodenkompetenz „Einsatz, Anwendung und Erzeugung von Wissen“

- Create visual narratives using charts and structured notebooks
- Organize and document analytical processes in alignment with CRISP-DM or KDD pipelines
- Interpret data in context using real-world use cases and small case studies

Selbstkompetenz „Wissenschaftliches Selbstverständnis und Professionalität“

- Reflect on learning development and decision-making under ambiguity
- Demonstrate initiative in managing and documenting your role within a project
- Develop awareness of bias and ethics in early-stage data exploration
- Learn from failure and feedback to iterate in the data process

Sozialkompetenz „Kommunikation und Kooperation“

- Collaborate in diverse project teams with defined roles
- Present insights to peers and instructors using clear visual and verbal communication
- Engage in peer-review, give and receive constructive feedback
- Participate actively in team-based problem-solving tasks

Constructive Alignment

- The module integrates the CRISP-DM and KDD data science process models to guide students through a structured approach to real-world data exploration, aligning theoretical understanding with applied project work.
- Lectures and impulse sessions introduce students to key concepts in data case framing, data cleaning, exploratory data analysis, and visualization, forming the conceptual basis for project execution.
- Through hands-on lab exercises, students acquire and apply technical skills in data wrangling, visualization, and pattern interpretation, which are directly applied within the context of the group project.
- The group project requires students to collaboratively analyze a real dataset, apply structured methodology, and develop a visual narrative—promoting specialist, methodological, and social competences.
- Peer-to-peer feedback, team reflection sessions, and rotating team roles foster social and self-competence development by encouraging collaboration, critical discussion, and awareness of learning contributions.
- The individual reflective report enables students to evaluate their learning experience, ethical considerations, and role in the data science process—supporting the development of personal and ethical awareness.
- All assessment components are aligned with competence-based learning objectives and structured according to the CORE educational model, ensuring coherence between intended outcomes, learning activities, and evaluation methods.

Inhalt des Moduls

1. Introduction to Data Science and the Data Case Approach

- What is data science? Overview of roles, processes, and impact
- Real-world use cases in business, health, logistics, and government
- Introduction to CRISP-DM and KDD as project frameworks
- Understanding "data cases" vs. predefined data tasks
- Problem framing: defining objectives, stakeholders, success criteria

2. Data Types, Formats, and Sources (Tool Focus: SAS Viya/Power BI/KNIME)

- Structured vs. unstructured data (tables, text, images)
- Common formats: CSV, Excel, SQL, JSON
- Accessing and loading data from files, databases, or public APIs
- Introduction to datasets and libraries
- Data import/export operations

3. Exploratory Data Analysis (EDA)

- Descriptive statistics: mean, median, mode, standard deviation
- Frequency tables and cross-tabulations
- Data visualization for exploration: histograms, bar charts, boxplots, scatter plots
- Identifying data quality issues (missing values, outliers, inconsistencies)

4. Data Preparation and Cleaning

- Handling missing data: deletion, imputation, flagging
- Recoding and transforming variables
- Data aggregation, filtering, sorting, merging
- Introduction to derived variables and calculated columns
- Data preparation workflows

5. Framing Data Science Questions

- From domain problem to data question
- Defining targets, predictors, and KPIs
- Building simple hypotheses and identifying assumptions
- Use cases in business, health, logistics, and media

6. Visual Storytelling and Insight Communication

- Data storytelling principles: clarity, purpose, and message
- Choosing the right visualization for comparisons, trends, distributions
- Building dashboards
- Filtering and interaction in reports
- Communicating results to decision-makers

Literatur

1. Kelleher, J. D., & Tierney, B. (2018). Data Science. MIT Press.
2. Martínez-Plumed, F., Contreras-Ochando, L., Ferri, C., & Hernández-Orallo, J. (2019). CRISP-DM Twenty Years Later: From Data Mining Processes to Data Science Trajectories. ACM Computing Surveys, 52(5), 1–39.
3. Fayyad, U., Piatetsky-Shapiro, G., & Smyth, P. (1996). From Data Mining to Knowledge Discovery in Databases. AI Magazine, 17(3), 37–54.

4. Tukey, J. W. (1977). Exploratory Data Analysis, Addison-Wesley.
5. Cairo, A. (2019). How Charts Lie: Getting Smarter about Visual Information., W. W. Norton & Company.
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Modulbeschreibung

Python Foundations for Data Science

Modulnummer	Niveau Master	Studiengangsleitung Prof. Dr. Swati Chandna
ECTS (Benotung) 6 (benotet)	Dauer Semester	Häufigkeit 2 per Year
Sprache English	Workload Insgesamt: 150 Stunden - davon Kontaktzeit: 34 Stunden - davon Selbststudium: 116 Stunden - davon Praxis: 0 Stunden	Veranstaltungstyp Lehrvortrag Übung

Teilnahmevoraussetzungen

Kursinformation

Modulprüfung

Prüfungsteile	Gewichtung (Bewertung)
Praktische Arbeit	100%

Qualifikationsziele (Learning Outcomes) und Kompetenzen

Nach erfolgreichem Abschluss des Moduls können die Studierenden ...

Handlungskompetenz

This module introduces students with little or no programming experience to the fundamentals of Python programming in the context of data science. It emphasizes practical coding, basic data analysis, and the use of essential libraries for handling and visualizing data. Students gain fluency in writing clear and functional code and applying it to real-world datasets in Jupyter notebooks.

Fachkompetenz „Wissen und Verstehen“

- Understand the structure and syntax of Python as a programming language.
- Differentiate between basic data types (int, float, str, bool) and complex types (lists, tuples, sets, dictionaries).
- Apply key programming constructs such as loops, conditionals, functions, and error handling.
- Use Python to automate data wrangling and exploratory analysis tasks.
- Work with structured data formats including CSV, JSON, and Excel using pandas.

Methodenkompetenz „Einsatz, Anwendung und Erzeugung von Wissen“

- Write clean, modular, and reusable Python code for data-related tasks.
- Use NumPy for numerical computation and array-based processing.
- Leverage pandas for tabular data manipulation, filtering, grouping, and aggregation.
- Visualize data effectively using matplotlib and seaborn for patterns and trends.
- Read, clean, and preprocess real-world datasets for further modeling or analysis.
- Structure and document code within Jupyter Notebooks for reproducibility.

Selbstkompetenz „Wissenschaftliches Selbstverständnis und Professionalität“

- Build confidence in solving programming challenges through iterative practice.
- Reflect on common programming pitfalls and personal debugging strategies.
- Demonstrate resilience and autonomy in troubleshooting errors and unexpected outputs.
- Develop a growth mindset by improving code quality and adapting learning tools (e.g., docs, Stack Overflow).
- Practice responsible coding habits, including commenting, documentation, and ethical data handling.

Sozialkompetenz „Kommunikation und Kooperation“

- Collaborate in coding pairs or small groups on mini-projects and debugging challenges.
- Share insights and strategies for solving problems with different approaches.
- Provide constructive code reviews and feedback in peer review sessions.
- Present and explain notebook outputs and plots to fellow learners.

Constructive Alignment

- Students first gain foundational programming skills through structured exercises, gradually applying them to real-world data problems.
- Core programming concepts are taught through interactive lab exercises that progress in complexity
- Each lab session builds toward a final portfolio, reinforcing method competence in data analysis with Python libraries.
- Coding challenges, pair programming, and peer feedback cultivate social competence and collaborative coding habits.

- Hands-on tutorials with real-world datasets help students understand the application of Python in data science.
- Peer activities (pair programming, sharing notebooks) build confidence and social skills in collaborative coding
- The final individual reflection emphasizes personal learning growth, confidence-building in Python, and problem-solving strategies.

Inhalt des Moduls

1. Introduction to Python and Programming Concepts

- Why Python for data science?
- The Python ecosystem: Python, pip, IDEs (Jupyter, VSCode, Google Colab)
- Variables, expressions, and assignment
- Basic data types: integers, floats, strings, booleans
- Type conversion and type() function
- Working in Jupyter Notebooks: markdown, code cells, outputs

2. Control Flow and Logic

- Conditional statements: if, elif, else
- Comparison and logical operators
- Loops: for, while, break, continue
- Looping through strings, lists, and ranges
- List comprehensions (basic form)

3. Python Data Structures

- Lists: indexing, slicing, appending, removing
- Tuples and immutability
- Dictionaries: key-value storage, iteration
- Sets: uniqueness and set operations
- Nesting data structures

4. Functions and Modular Programming

- Defining and calling functions
- Parameters, arguments, return values
- Variable scope and default arguments
- Writing reusable, modular code
- Lambda expressions (intro level)

5. Working with Files and Input/Output

- File paths and directories
- Reading from and writing to text/CSV files using open()
- Basic error handling with try/except
- Using with statements and file iteration
- Parsing simple data from structured files

6. Python and Data with pandas and NumPy

- What are pandas and NumPy?
- Series and DataFrames: loading, viewing, summarizing
- Selecting, filtering, and slicing data
- Applying basic statistics (mean, median, value_counts)
- Sorting and grouping
- Basic array manipulations with NumPy

7. Data Visualization Basics

- Why visualize data? Principles of clear plotting
- Introduction to matplotlib and seaborn
- Plot types: line, bar, scatter, histogram, boxplot
- Customizing titles, labels, colors, legends
- Saving visualizations to files

8. Working with External Libraries and APIs

- Installing packages with pip
- Importing and using third-party libraries
- Introduction to JSON and APIs
- Making simple API calls with requests
- Parsing JSON responses into dictionaries

9. Debugging, Testing, and Writing Clean Code

- Common errors and how to interpret tracebacks
- Print-based debugging vs. using pdb
- Writing simple assertions and tests
- PEP8 and good naming conventions
- Commenting and documenting code

Literatur

- Martin RC: The Clean Coder: A Code of Conduct for Professional Programmers, 1st edition, Prentice Hall, 2011.
- McKinney, W. (2022). Python for Data Analysis (3rd ed.).
- VanderPlas, J. (2016). Python Data Science Handbook.
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- Martin RC: Clean Architecture: A Craftsman's Guide to Software Structure and Design, 1st edition, Prentice Hall, 2017.
- Ramalho, L: Fluent Python, 1st edition, O'Reilly, 2015
- Wickham H & Golemund G, R for Data Science, 1st edition, O'Reilly, 2017
- McKinney, Wes. Python for Data Analysis: Data Wrangling with Pandas, NumPy, and IPython. O'Reilly Media, 2017.
- Grus, Joel. Data Science from Scratch: First Principles with Python. O'Reilly Media, 2015.
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- Albon, Chris. Machine Learning with Python Cookbook: Practical Solutions from Preprocessing to Deep Learning. O'Reilly Media, 2018.
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Modulbeschreibung

Data Engineering

Modulnummer	Niveau Master	Studiengangsleitung Prof. Dr. Swati Chandna
ECTS (Benotung) 6 (benotet)	Dauer Semester	Häufigkeit 2 per Year
Sprache English	Workload Insgesamt: 150 Stunden - davon Kontaktzeit: 45 Stunden - davon Selbststudium: 105 Stunden - davon Praxis: 0 Stunden	Veranstaltungstyp Gruppenarbeit Lehrvortrag Tutorium Übung

Teilnahmevoraussetzungen

Kursinformation

Modulprüfung

Prüfungsteile	Gewichtung (Bewertung)
Projektarbeit	100%

Qualifikationsziele (Learning Outcomes) und Kompetenzen

Nach erfolgreichem Abschluss des Moduls können die Studierenden ...

Handlungskompetenz

This module introduces the fundamental building blocks of modern data engineering. Students learn to design, query, manage, and extract data from both structured and unstructured sources, interact with relational and non-relational databases, and understand the data infrastructure underpinning scalable systems, including Hadoop and cloud platforms.

Fachkompetenz „Wissen und Verstehen“

Graduates of this module will possess the following knowledge and understanding:

- Define and explain the core concepts, lifecycle, and role of data engineering within the broader data landscape.
- Describe and differentiate between components of modern data infrastructure, including Big Data systems, cloud platforms, and data pipelines.
- Understand the Hadoop ecosystem and cloud computing models (IaaS, PaaS, SaaS), including their advantages and disadvantages.
- Differentiate between various data storage solutions, including distributed file systems, RAID, relational, and NoSQL databases.
- Explain the concepts, techniques, and ethical considerations of web scraping for data acquisition.
- Understand the relational database model, architecture, and be proficient in using SQL for data definition and manipulation.
- Describe the principles and use cases of different NoSQL database types (document, key-value, wide-column, graph).
- Understand and apply concepts of data partitioning, replication, and availability in distributed systems.

Methodenkompetenz „Einsatz, Anwendung und Erzeugung von Wissen“

After completing this module, students will have developed the ability to:

- Design and implement data pipelines for ingesting, processing, and storing data from various sources.
- Apply appropriate tools and technologies for specific data engineering tasks, such as Hadoop, Spark, cloud-based services (e.g., AWS), and database systems.
- Write SQL queries to extract, transform, and load data in relational databases.
- Utilize NoSQL databases for specific data storage and retrieval needs, considering their strengths and limitations.
- Develop web scraping scripts to collect data from websites, adhering to ethical guidelines and legal regulations.
- Evaluate and select appropriate data infrastructure components based on specific project requirements and constraints.
- Troubleshoot and resolve issues related to data infrastructure and pipelines.
- Analyze and interpret data generated from various data systems.

Selbstkompetenz „Wissenschaftliches Selbstverständnis und Professionalität“

This module aims to cultivate the following aspects of self-competence in students:

- Independently research and evaluate new data engineering technologies and trends.
- Critically assess the ethical implications of data collection, storage, and processing.
- Manage time and resources effectively when working on data engineering projects.
- Adapt to evolving technologies and methodologies in the field of data engineering.
- Take ownership of project deliverables and demonstrate accountability for their work.

Sozialkompetenz „Kommunikation und Kooperation“

The activities and assessments within this module are designed to enhance students' social skills, enabling them to:

- Collaborate effectively with team members on data engineering projects, sharing knowledge and expertise.
- Communicate technical concepts and solutions clearly and concisely to both technical and non-technical audiences.
- Present project findings and recommendations in a professional manner.
- Participate constructively in discussions and provide feedback to peers.
- Negotiate and resolve conflicts that may arise during collaborative project work.

Constructive Alignment

The learning activities, course content, and assessment in this module are aligned to achieve the stated learning outcomes. The module employs a project-based learning approach, where students will work on a substantial data engineering project throughout the semester. This project will require students to apply the theoretical concepts and practical skills learned in lectures and workshops to a real-world scenario. The "Project Work" assessment will evaluate the students' ability to:

- Design: Conceptualize a data infrastructure solution that meets specific requirements.
- Implement: Build a working prototype of the designed solution, using appropriate tools and technologies.
- Document: Clearly document the project's design, implementation, and results.
- Present: Effectively communicate the project's findings and demonstrate the functionality of the solution.
- Reflect: Critically reflect on the project and identify challenges.

Inhalt des Moduls

1. Introduction to Data Engineering

- Role of data engineering in the data lifecycle
- Differences between data science, data analytics, and data engineering
- Overview of the data pipeline: ingestion → transformation → storage → access
- Real-world applications: ETL, analytics engineering, ML pipelines
- Key job roles and tech stack in modern data engineering

2. Data Infrastructure and Architecture

- Overview of modern data infrastructure: data lakes, warehouses, lakehouses
- Batch vs. real-time data processing
- Distributed systems overview: horizontal vs. vertical scaling
- Introduction to data orchestration (e.g. Airflow, Prefect - conceptual)
- Architecture drawing exercises

3. Hadoop Ecosystem and Distributed Data Processing

- Hadoop: HDFS, YARN, MapReduce
- Introduction to Spark and Hive
- File formats: CSV, JSON, Avro, Parquet, ORC
- Compression and partitioning for storage efficiency
- Differences between row- and column-based storage
- Spark SQL and PySpark examples (basic level)

4. Cloud Computing for Data Engineering

- Cloud models: IaaS, PaaS, SaaS
- Common cloud platforms: AWS, Azure, Google Cloud (GCP)
- Overview of data services:
 - AWS: S3, Redshift, Glue
 - GCP: BigQuery, Cloud Storage
 - Azure: Synapse, Blob Storage
- Introduction to cloud-native data pipelines
- Sample access to cloud console (e.g., Google BigQuery sandbox or AWS Free Tier)

5. Data Storage and File Management

- Local storage vs. distributed file systems
- Object storage, block storage, and databases
- Principles of durable, fault-tolerant storage systems
- File organization, naming conventions, metadata management
- Working with public data in cloud storage (e.g., Kaggle Datasets on GCS)

6. Web Scraping and Data Ingestion

- Overview of data ingestion: batch, stream, and API-based
- Introduction to REST APIs, rate limits, JSON response parsing
- Basics of HTML structure and web crawling ethics
- Python for scraping: requests, BeautifulSoup, Selenium

7. Databases and Querying Fundamentals

- What is a DBMS? OLTP vs. OLAP systems
- Normalization, ER diagrams, schemas
- Indexing and query optimization
- Differences between relational and non-relational databases

8. Relational Databases and SQL

- SQL syntax: SELECT, WHERE, JOIN, GROUP BY, ORDER BY
- Nested queries, aggregate functions
- CASE statements, views, stored procedures (conceptual overview)
- Transaction handling and ACID properties

9. Advanced SQL and Performance Tuning

- Window functions and analytic queries
- Common Table Expressions (CTEs) and recursive queries
- Query performance optimization and indexing
- Query plans and execution time diagnostics

10. NoSQL and Schema-less Databases

- Introduction to NoSQL types: document, key-value, wide-column, graph
- Use cases for MongoDB, Redis, Cassandra, Neo4j
- CAP theorem and BASE vs. ACID
- Indexing and performance in NoSQL stores

11. Wide-Column Databases and Big Data Storage

- Introduction to Cassandra, HBase, and Google Bigtable
- Columnar storage model and scalability
- Partitioning, replication, and consistency models
- Schema design for large-scale application
- Relational Databases: Entity-relationship models, normalization, ACID principles
- Basic SQL Queries: SELECT, WHERE, JOIN, GROUP BY
- Advanced SQL Techniques: Window functions, subqueries, indexing, views
- NoSQL Systems: Document (MongoDB), Key-value (Redis), Graph (Neo4j)
- Wide-column Databases: Cassandra, Google Bigtable concepts and use cases

Literatur

- Kimball, R., & Ross, M. (2013). The Data Warehouse Toolkit: The Definitive Guide to Dimensional Modeling (3rd ed.). Wiley.
- White, T. (2015). Hadoop: The Definitive Guide (4th ed.). O'Reilly Media.
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- Ramakrishnan, R., & Gehrke, J. (2003). Database Management Systems. McGraw-Hill Education.
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- Celko, J. (2013) Joe Celko's SQL for Smarties: Advanced SQL Programming. Morgan Kaufmann

Modulbeschreibung

Statistics and Machine Learning

Modulnummer	Niveau Master	Studiengangsleitung Prof. Dr. Swati Chandna
ECTS (Benotung) 6 (benotet)	Dauer Semester	Häufigkeit 2 per Year
Sprache English	Workload Insgesamt: 150 Stunden - davon Kontaktzeit: 64 Stunden - davon Selbststudium: 86 Stunden - davon Praxis: 0 Stunden	Veranstaltungstyp Gruppenarbeit Lehrvortrag

Teilnahmevoraussetzungen

Kursinformation

Modulprüfung

Prüfungsteile	Gewichtung (Bewertung)
Projektarbeit	100%

Qualifikationsziele (Learning Outcomes) und Kompetenzen

Nach erfolgreichem Abschluss des Moduls können die Studierenden ...

Handlungskompetenz

Students will build a strong foundation and practical mastery in statistics and machine learning, enabling them to analyze, model, and predict outcomes from real-world datasets with precision. The goal is to equip learners with robust Python-based skills in data handling and ML algorithms, fostering critical, data-driven decision-making across diverse fields like business, science, and media. Through applied projects, they will develop the ability to tackle complex problems and adapt to evolving analytical challenges.

Fachkompetenz „Wissen und Verstehen“

- Distinguish between descriptive and inferential statistical techniques and identify their appropriate use cases.
- Understand probability distributions (Normal, Binomial, Poisson) and their role in modeling uncertainty.
- Apply core statistical tools (e.g., hypothesis testing, correlation analysis, linear regression) to draw conclusions from data.
- Explain the taxonomy and key concepts of machine learning, including supervised, unsupervised, and time series learning.
- Understand the assumptions and appropriate applications of ARIMA and LSTM for time series forecasting.
- Differentiate between parametric and non-parametric models and evaluate trade-offs.
- Articulate the role of feature selection and engineering in influencing model performance.

Methodenkompetenz „Einsatz, Anwendung und Erzeugung von Wissen“

- Conduct statistical analysis and hypothesis testing using Python libraries such as scipy.stats, statsmodels, and pandas.
- Build and evaluate machine learning models (regression, classification, time series) using scikit-learn, statsmodels, and keras.
- Implement cross-validation techniques and performance metrics (e.g., F1-score, ROC AUC, RMSE, MAE) to assess models rigorously.
- Preprocess data for ML pipelines including handling of missing data, scaling, encoding, and outlier treatment.
- Apply ARIMA models using statsmodels and construct basic LSTM models using Keras for sequence prediction tasks.
- Use Jupyter Notebooks to document, visualize, and present analytical workflows in a reproducible format.
- Perform feature importance analysis and model diagnostics (e.g., residual plots, overfitting checks).

Selbstkompetenz „Wissenschaftliches Selbstverständnis und Professionalität“

- Reflect on the limitations, assumptions, and ethical boundaries of statistical inference and machine learning applications.
- Demonstrate independent problem-solving in debugging code, tuning models, or interpreting ambiguous results.
- Acknowledge uncertainty and variation in real-world data and avoid overgeneralizing model outcomes.
- Develop personal strategies for handling the iterative nature of data science projects, including learning from failed experiments.
- Recognize the potential societal impact of models, particularly in domains like healthcare, finance, and criminal justice.
- Articulate personal learning growth and contributions through reflective journaling or oral assessments.
- Make responsible choices when accessing, storing, or sharing datasets (e.g., handling personally identifiable information).

Sozialkompetenz „Kommunikation und Kooperation“

- Collaborate in interdisciplinary teams to define a problem, analyze data, and communicate insights through storytelling.
- Take on varied project roles (e.g., modeler, data wrangler, presenter, documenter) and coordinate work distribution.
- Communicate technical results clearly to non-technical stakeholders using visuals and intuitive summaries.
- Participate in group discussions to critically evaluate different modeling strategies and collectively make decisions.
- Provide and incorporate peer feedback during model development and presentation rehearsals.
- Demonstrate sensitivity to ethical and social implications when choosing and communicating models (e.g., bias in classification models).

Constructive Alignment

Through lectures, guided exercises, and team-based projects, students connect theoretical statistical knowledge with hands-on ML implementation. Emphasis is placed on understanding and applying data modeling, using tools aligned with current industry practices, and reflecting on outcomes to iterate and improve solutions.

- Statistical foundations are taught through real-world datasets and applied using Python libraries, ensuring method and specialist competence.
- Machine learning models are introduced through live examples and applied in small-scale lab tasks and a larger group project.
- Time series analysis is practiced through step-by-step labs on real forecasting problems (e.g., electricity consumption or COVID-19 cases).
- The group project integrates the full data pipeline: data exploration, preprocessing, modeling, evaluation, and communication.
- All assessments support iterative learning and align with CORE's emphasis on competence-oriented, practice-based education.

Inhalt des Moduls

1. Statistics and Probability

- Descriptive vs. Inferential Statistics
- Measures of central tendency and dispersion
- Data types and visual exploration
- Probability, Sampling & Central Limit Theorem
- Sampling techniques, distributions (Normal, Binomial, Poisson)
- Central limit theorem implications
- Hypothesis Testing, Correlation, and Regression
- t-tests, ANOVA, chi-square tests
- Pearson/Spearman correlation
- Simple and multiple linear regression

2. Machine Learning

- Introduction to ML
- Types: Supervised, Unsupervised, Reinforcement
- Applications in various domains (health, finance, marketing)
- Data Preprocessing
- Handling missing data, encoding, scaling
- Feature selection and engineering techniques

3. Modeling Techniques

- Regression: linear, ridge, lasso
- Classification: logistic regression, decision trees, random forest, SVM, Naïve Bayes

4. Time Series Forecasting:

- Concepts: stationarity, seasonality
- Models: ARIMA, SARIMA, introduction to LSTM with Keras

5. Model Evaluation

- Train/test split, cross-validation
- Confusion matrix, precision, recall, F1-score
- RMSE, MAE for regression/forecasting models

Literatur

- Diez, D., Barr, C., & Çetinkaya-Rundel, M. (2019). OpenIntro Statistics.
- James, G., Witten, D., Hastie, T., & Tibshirani, R. (2021). An Introduction to Statistical Learning. Springer.
- Géron, A. (2019). Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow. O'Reilly Media.
- Raschka, S., & Mirjalili, V. (2020). Python Machine Learning. Packt Publishing.
- Hyndman, R. J., & Athanasopoulos, G. (2021). Forecasting: Principles and Practice (3rd ed.).



Semester 1: General or Interdisciplinary Track

Campus Specific



General Track

Heidelberg

Modulbeschreibung

Advanced Python for Data Science

Modulnummer	Niveau Master	Studiengangsleitung Prof. Dr. Swati Chandna
ECTS (Benotung) 6 (benotet)	Dauer Semester	Häufigkeit 2 per Year
Sprache English	Workload Insgesamt: 150 Stunden - davon Kontaktzeit: 34 Stunden - davon Selbststudium: 116 Stunden - davon Praxis: 0 Stunden	Veranstaltungstyp Lehrvortrag Übung

Teilnahmevoraussetzungen

Kursinformation

Modulprüfung

Prüfungsteile	Gewichtung (Bewertung)
Praktische Arbeit	100%

Qualifikationsziele (Learning Outcomes) und Kompetenzen

Nach erfolgreichem Abschluss des Moduls können die Studierenden ...

Handlungskompetenz

This module enables students to develop production-grade, scalable, and high-performance Python applications for data science, engineering, and AI use cases. Learners explore advanced programming concepts, asynchronous and concurrent computation, containerization, cloud-native deployment, and GPU acceleration using CUDA, CuPy, and TensorFlow/PyTorch integrations.

Fachkompetenz „Wissen und Verstehen“

- Explain advanced Python features: decorators, context managers, metaclasses, async/await
- Describe GPU architecture and CUDA programming principles
- Understand how Python interacts with CUDA via libraries like CuPy and Numba
- Evaluate performance trade-offs between CPU and GPU execution in data-intensive tasks
- Understand architectural components of cloud deployments and container-based services

Methodenkompetenz „Einsatz, Anwendung und Erzeugung von Wissen“

- Implement concurrent and asynchronous Python workflows using asyncio, threading, and multiprocessing
- Build and test Flask APIs and Python packages for deployment
- Use Docker, Kubernetes, and CI/CD pipelines for scalable deployments
- Perform GPU-accelerated computations using CuPy, Numba, or TensorFlow
- Profile and optimize Python code for performance (e.g., memory usage, computation time)

Selbstkompetenz „Wissenschaftliches Selbstverständnis und Professionalität“

- Troubleshoot multi-layered issues in Python ecosystems (e.g., concurrency bugs, CUDA compatibility)
- Evaluate ethical and environmental considerations of GPU-based computation (e.g., energy impact)
- Reflect on tool selection and design decisions within real-world constraints
- Demonstrate adaptability in integrating new libraries or deploying across cloud/GPU environments

Sozialkompetenz „Kommunikation und Kooperation“

- Work collaboratively on a Git-managed project with CI pipelines
- Practice agile development: planning, versioning, pull requests, and code reviews
- Communicate performance analysis, optimization decisions, and deployment strategies clearly
- Document architecture and coding workflows for reuse and reproducibility

Constructive Alignment

- Students begin with complex Python concepts and apply them to web development, API design, and container-based deployment.
- Labs introduce concurrency and GPU programming, culminating in a team project that applies real-time data processing or ML pipelines with GPU acceleration.
- Docker, CI/CD, and Kubernetes workflows support professional software engineering and cloud readiness.
- Peer reviews, code documentation, and oral reflection tasks enhance personal and social competence.

Inhalt des Moduls

- Advanced Python Programming: Generators, decorators, context managers, Functional programming constructs
- Concurrency and Asynchronous Workflows: Multi-threading, multi-processing, Asynchronous I/O with asyncio, aiohttp, Task queues and scheduling basics
- GPU Programming and Performance: Introduction to CUDA and GPU architecture, GPU profiling and memory management
- Software Design and Testing: Clean code and modular design (PEP8, SOLID), Test-driven development: pytest, mocking, CI integration, Documentation: docstrings, Sphinx, Swagger for APIs
- Web Development and API Design: Building RESTful APIs with Flask, API structure, authentication, deployment options, Dockerization of Flask apps with GPU support
- Cloud Deployment and DevOps for Data Science: Cloud service models (IaaS, PaaS, SaaS), CI/CD with GitHub Actions or GitLab CI, Container orchestration and GPU scheduling in Kubernetes
-

Literatur

- Luciano Ramalho. (2021). Fluent Python (2nd ed.)
- Anderson, J. et al. (2021). CUDA by Example: An Introduction to General-Purpose GPU Programming.
- Raschka, S. & Mirjalili, V. (2022). Python Machine Learning (4th ed.)
- Grinberg, M. (2018). Flask Web Development (2nd ed.)
- Goldwasser, M. et al. (2020). Object-Oriented Programming in Python.



Creative AI and Media Analytics

Heidelberg

Modulbeschreibung

Recommendation Systems and Media Personalization

Modulnummer	Niveau Master	Studiengangsleitung Prof. Dr. Swati Chandna
ECTS (Benotung) 6 (benotet)	Dauer Semester	Häufigkeit 2 per Year
Sprache English	Workload Insgesamt: 150 Stunden - davon Kontaktzeit: 34 Stunden - davon Selbststudium: 116 Stunden - davon Praxis: 0 Stunden	Veranstaltungstyp Gruppenarbeit Lehrvortrag

Teilnahmevoraussetzungen

Kursinformation

Modulprüfung

Prüfungsteile	Gewichtung (Bewertung)
Projektarbeit	100%

Qualifikationsziele (Learning Outcomes) und Kompetenzen

Nach erfolgreichem Abschluss des Moduls können die Studierenden ...

Handlungskompetenz

Students will develop comprehensive competencies in designing, implementing, and evaluating recommender systems and personalization techniques using advanced algorithms and state-of-the-art tools. Learners will build interdisciplinary skills combining technical expertise, user-focused design, and ethical considerations within diverse data-driven scenarios. Through practical exploration of real-world datasets, students will develop adaptive systems that enhance user experience, effectively addressing personalization challenges, and preparing them for professional roles in sectors such as digital media, e-commerce, and personalized content delivery. This module introduces students to the theoretical foundations, algorithms, and practical applications of recommendation systems. It explores collaborative filtering, content-based filtering, hybrid methods, and contextual personalization techniques as applied in media, retail, entertainment, and social platforms. Students build and evaluate their own recommendation engines using real-world datasets, critically analyzing their performance, fairness, and personalization strategies.

Fachkompetenz „Wissen und Verstehen“

- Understand the taxonomy of recommendation systems (e.g., collaborative, content-based, hybrid)
- Explain user-item interaction modeling using matrix factorization, nearest neighbor, and deep learning
- Interpret evaluation metrics: precision, recall, MAP, NDCG, diversity, serendipity
- Analyze the role of personalization in digital platforms (e.g., media, e-commerce, news)

Methodenkompetenz „Einsatz, Anwendung und Erzeugung von Wissen“

- Implement basic recommendation models using Python (e.g., cosine similarity, matrix factorization with surprise or lightFM)
- Use libraries such as scikit-learn, surprise, implicit, lightFM, or TensorFlow/Keras (optional)
- Perform offline evaluation with cross-validation and top-N ranking metrics
- Apply preprocessing techniques for sparse data, cold-start problems, and implicit feedback

Selbstkompetenz „Wissenschaftliches Selbstverständnis und Professionalität“

- Reflect critically on the limitations of algorithmic personalization
- Make informed decisions about the trade-offs between accuracy, diversity, and explainability
- Demonstrate initiative in adapting recommendation models to new domains or datasets
- Address ethical considerations in data collection and personalized content delivery

Sozialkompetenz „Kommunikation und Kooperation“

- Collaborate in teams to design, develop, and evaluate recommender systems for real-world scenarios
- Translate algorithmic output into user-facing experiences using visualizations or UI mockups
- Communicate model choices and evaluation outcomes to non-technical stakeholders
- Engage in debates around filter bubbles, privacy, bias, and fairness in personalization systems

Constructive Alignment

Students will apply their learning through a team-based project that mirrors real-world applications. They design, implement, and evaluate a recommender system or personalization solution using actual datasets. Future skills such as critical thinking, digital literacy, collaboration, and adaptive learning are embedded through project iterations, peer interaction, and final presentations.

- Lectures and labs introduce models and metrics, grounded in theory and implemented using Python toolkits
- Group projects allow students to choose a real-world domain (e.g., movies, music, news) and design a

recommendation pipeline

- Discussions and presentations foster understanding of ethical trade-offs and design choices in personalization systems
- Reflection reports reinforce self-awareness in evaluating performance, transparency, and social responsibility
- Activities align with CORE's emphasis on applied, collaborative, and ethical competence development

Inhalt des Moduls

1. Introduction to Recommender Systems and Personalization

- What are recommender systems?
- Types of recommendations: personalized vs. non-personalized
- Use cases across industries: e-commerce, media, healthcare, education, travel
- Overview of user modeling and personalization strategies
- Real-world examples: Netflix, Amazon, Spotify, TikTok, LinkedIn

2. Collaborative Filtering: User-Based and Item-Based

- Memory-based collaborative filtering
- User-user similarity
- Item-item similarity
- Similarity metrics: cosine similarity, Pearson correlation
- Cold start, sparsity, and scalability challenges

3. Content-Based Filtering and Hybrid Approaches

- User profiles and item metadata
- TF-IDF, bag-of-words models, cosine similarity on content features
- Hybrid recommenders: weighted, switching, cascade, feature-augmented
- Combining behavioral and content signals

4. Implicit and Explicit Feedback Mechanisms

- Definitions and differences
- Explicit: ratings, likes, reviews
- Implicit: clicks, views, dwell time, purchases
- Weighting strategies for implicit data
- Confidence modeling in implicit data (Hu, Koren, Volinsky, 2008)

5. Matrix Factorization and Deep Learning for Recommendations

- Model-based CF:
- Singular Value Decomposition (SVD), SVD++
- Non-negative Matrix Factorization (NMF)
- Neural Collaborative Filtering (NCF)
- Embedding users/items
- TensorFlow Recommenders and PyTorch implementations

6. Context-Aware and Sequential Recommendations

- Contextual features: location, time, device, mood
- Approaches: contextual pre-filtering, post-filtering, model-based
- Sequential recommenders:
- Markov Chains

7. Evaluation Metrics for Recommendation and Personalization

- Offline vs. online evaluation
- Ranking-based metrics: precision@k, recall@k, NDCG, MAP
- Coverage, novelty, diversity, serendipity
- Introduction to A/B testing and online experimentation (conceptual)

- Metric trade-offs in real-world deployments

8. Bias, Fairness, and Ethics in Recommender Systems

- Sources of bias: popularity bias, exposure bias, selection bias
- Fairness in personalization (e.g., user group balance)
- Filter bubbles and echo chambers
- GDPR, consent, and explainability requirements
- Techniques for mitigating bias (re-ranking, diversity filters, fairness metrics)

Literatur

- Ricci, F., Rokach, L., & Shapira, B. (2015). Recommender Systems Handbook. Springer.
- Aggarwal, C. C. (2016). Recommender Systems: The Textbook. Springer.
- O'Reilly Media. (2020). Practical Recommender Systems.
- Grus, J. (2015). Data Science from Scratch. O'Reilly Media.
- McKinney, W. (2017). Python for Data Analysis. O'Reilly Media.



Supply Chain and Logistics Analytics

Hamburg

Modulbeschreibung

Demand Forecasting

Modulnummer	Niveau Master	Studiengangsleitung Prof. Dr. Swati Chandna
ECTS (Benotung) 6 (benotet)	Dauer Semester	Häufigkeit 2 per Year
Sprache English	Workload Insgesamt: 150 Stunden - davon Kontaktzeit: 34 Stunden - davon Selbststudium: 116 Stunden - davon Praxis: 0 Stunden	Veranstaltungstyp Lehrvortrag Übung

Teilnahmevoraussetzungen

Kursinformation

Modulprüfung

Prüfungsteile	Gewichtung (Bewertung)
Projektarbeit	100%

Qualifikationsziele (Learning Outcomes) und Kompetenzen

Nach erfolgreichem Abschluss des Moduls können die Studierenden ...

Handlungskompetenz

The goal of this course is to equip students with the essential knowledge and skills to effectively apply data analytics and optimization techniques in the management and optimization of supply chains. By understanding the foundational principles of Supply Chain Management (SCM), the importance of data-driven decision-making, and advanced analytical methods, students will be prepared to enhance supply chain efficiency, reduce costs, improve customer satisfaction, and mitigate risks. Through hands-on learning of predictive analytics, demand forecasting, supply chain risk management, and transportation optimization, students will gain practical expertise in leveraging cutting-edge data science tools, algorithms, and models to address complex supply chain challenges in real-world scenarios.

Fachkompetenz „Wissen und Verstehen“

- Possess specialized competence in applying data science and analytics to optimize supply chain operations.
- Understand key supply chain principles, including demand planning, inventory turnover, lead time variability, and service levels.
- Forecast demand using advanced techniques such as time series decomposition, ARIMA, regression-based models, and machine learning-based forecasting.
- Evaluate and improve performance using forecasting KPIs such as MAE, RMSE, MAPE, and WAPE, and align forecasts with business objectives.
- Conduct scenario and sentiment analysis to model uncertainty, demand shocks, and market dynamics.
- Apply predictive and prescriptive analytics to real-world problems, integrating external drivers (e.g., promotions, weather, economic indicators) into models.
- Understand and apply optimization algorithms (e.g., linear programming, greedy heuristics) for cost reduction, transportation scheduling, and route planning.
- Support product lifecycle planning by analyzing trends, demand cycles, and market saturation effects.
- Analyze demand volatility and seasonality to support strategic decisions in procurement, production, and distribution.
- Communicate insights from forecasting models to cross-functional stakeholders in operations, finance, and sales using dashboards and simulation tools.

Methodenkompetenz „Einsatz, Anwendung und Erzeugung von Wissen“

- Preprocess time series data: stationarity, seasonality, trend decomposition
- Implement univariate and multivariate models using Python (statsmodels, scikit-learn, XGBoost, Prophet)
- Apply and evaluate time series cross-validation techniques
- Quantify forecast accuracy using MAE, MAPE, RMSE, SMAPE, WAPE
- Use feature engineering for external variables (promotions, weather, holidays)

Selbstkompetenz „Wissenschaftliches Selbstverständnis und Professionalität“

- Reflect on model limitations, uncertainty, and ethical consequences of forecasts
- Make informed modeling and preprocessing decisions in ambiguous business contexts
- Demonstrate ownership of modeling tasks and iterative improvement
- Show accountability in delivering business-relevant results under time constraints

Sozialkompetenz „Kommunikation und Kooperation“

- Work in teams to build and present forecasting solutions for real datasets
- Interpret and communicate forecast results to both technical and business audiences

- Collaborate with domain-specific experts to understand use case requirements
- Discuss risk and uncertainty around forecast scenarios with decision-makers

Constructive Alignment

- This module is designed to equip students with the knowledge and skills to optimize supply chain operations using data analytics and advanced optimization techniques.
- Lectures and readings provide the theoretical foundation in demand forecasting, supply chain principles, and predictive analytics.
- Hands-on lab sessions and software tools enable students to apply time series models, regression techniques, and simulation approaches to real-world datasets.
- Case studies and group projects encourage collaborative problem-solving and the application of forecasting models to challenges such as product lifecycle planning, transportation routing, and cost reduction.
- Exercises and project presentations assess students' ability to analyze data, select appropriate forecasting and optimization techniques, and communicate insights effectively to business stakeholders.
- Students learn to manage risks through scenario analysis and sentiment-informed forecasting and to align forecasts with KPIs and supply chain performance metrics.
- The integration of simulation and prescriptive analytics ensures that students develop actionable recommendations to address complex supply chain issues.
- By linking course activities to real-world applications, this alignment ensures that students gain the expertise necessary to make data-driven decisions and design resilient, efficient supply chain strategies.

Inhalt des Moduls

1. Introduction to Supply Chain Management and Analytics

- What is a supply chain? Flows of materials, information, and finances
- Evolution of supply chains: from linear to digital networks
- Key drivers: facilities, inventory, transportation, information, sourcing, pricing

2. Introduction to Demand Forecasting

- What is demand forecasting and why is it critical?
- Applications in retail, logistics, energy, and healthcare
- Types of demand: cyclical, seasonal, intermittent, declining
- Levels of forecasting: strategic, tactical, operational
- Overview of demand planning systems and organizational context

3. Forecasting Frameworks and Methodologies

- Introduction to forecasting process models:
- CRISP-DM for demand prediction workflows
- KDD and business understanding in forecasting
- Framing forecasting problems: objectives, KPIs, constraints
- Exploratory Data Analysis (EDA) for time series

4. Classical Time Series Forecasting Methods

- Naive and seasonal naive forecasting
- Moving average, exponential smoothing (SES, Holt, Holt-Winters)
- Decomposition: trend, seasonality, residual (additive vs multiplicative)
- Stationarity and differencing
- ARIMA, SARIMA, and auto.arima (AIC/BIC selection)

5. Feature-Based and Machine Learning Forecasting

- Feature engineering from time (day of week, holiday, lag features)
- Regression-based forecasting (linear, ridge, lasso)
- Decision trees and ensemble methods (Random Forest, XGBoost)
- Recursive vs. direct multi-step forecasting

6. Forecast Evaluation and Model Selection

- Error metrics: MAE, MAPE, RMSE, WAPE, SMAPE
- Cross-validation for time series: rolling/expanding windows
- Business-centric evaluation: forecast bias, service levels, inventory impact
- Benchmarking baseline vs. advanced models

Literatur

- Vandeput, N. (2021). Data Science for Supply Chain Forecasting. Berlin, Boston: De Gruyter.
<https://doi.org/10.1515/9783110671124>
- Vandeput, N. (2023). Demand Forecasting Best Practices. Manning Publications.
- Chopra, S., & Meindl, P. (2019). Supply Chain Management: Strategy, Planning, and Operation (7th ed.). Pearson Education.
- Ivanov, D. (2024). Introduction to Supply Chain Analytics Classroom Companion: Business.
- Hyndman, R. J., & Athanasopoulos, G. (2021). Forecasting: Principles and Practice (3rd ed.)

- Montgomery, D. C., Jennings, C. L., & Kulahci, M. (2015). Introduction to Time Series Analysis and Forecasting



Business Analytics

Fürth

Modulbeschreibung

Trend Analysis and Consumer Behaviour

Modulnummer	Niveau Master	Studiengangsleitung Prof. Dr. Clemens Werkmeister
ECTS (Benotung) 6 (benotet)	Dauer Semester	Häufigkeit 2 per Year
Sprache English	Workload Insgesamt: 150 Stunden - davon Kontaktzeit: 34 Stunden - davon Selbststudium: 116 Stunden - davon Praxis: 0 Stunden	Veranstaltungstyp Lehrvortrag Übung Gruppenarbeit

Teilnahmevoraussetzungen

Kursinformation

Modulprüfung

Prüfungsteile	Gewichtung (Bewertung)
Studienarbeit	100%

Qualifikationsziele (Learning Outcomes) und Kompetenzen

Nach erfolgreichem Abschluss des Moduls können die Studierenden ...

Handlungskompetenz

Students will be able to analyze and interpret emerging market trends and patterns in consumer behavior using data-driven methods. They will learn to translate insights into strategic recommendations for business decision-making in dynamic market environments.

Fachkompetenz „Wissen und Verstehen“

Students understand the fundamental theories and concepts of consumer behavior and their relevance in modern business contexts. They are able to apply trend analysis techniques to real-world data to identify shifts in consumer preferences and market dynamics.

Methodenkompetenz „Einsatz, Anwendung und Erzeugung von Wissen“

Students acquire the ability to utilize statistical, machine learning, and data visualization techniques for identifying and interpreting trends. They learn to design and conduct consumer research using structured data analysis frameworks.

Selbstkompetenz „Wissenschaftliches Selbstverständnis und Professionalität“

Students develop skills in presenting analytical findings to diverse stakeholders and collaborating effectively in interdisciplinary teams. They engage in respectful discourse, considering multiple perspectives in group decision-making.

Sozialkompetenz „Kommunikation und Kooperation“

Students enhance their capacity for critical thinking and reflection when evaluating market information. They take ownership of complex analysis tasks, demonstrating responsibility and initiative in their learning and application.

Constructive Alignment

Learning objectives are achieved through the integration of theoretical inputs, hands-on case studies, and data analysis projects. Assessment tasks (e.g., project reports and presentations) are aligned with real-world scenarios requiring students to apply analytical methods to consumer datasets and deliver actionable business insights.

Inhalt des Moduls

- Introduction to consumer behavior theories and models
- Fundamentals of trend analysis in a business context
- Data sources for consumer and market research
- Techniques for identifying market and behavioral trends (e.g., time series analysis, clustering, sentiment analysis)
- Behavioral economics and digital consumer behavior
- Use of Python/R for consumer data analysis
- Data storytelling and visualization of trends
- Case studies on real market and consumer data
- Ethical considerations in trend prediction and consumer profiling

Presentation of insights to stakeholders

Literatur

- Solomon, M. R. (2019). Consumer Behavior: Buying, Having, and Being (13th ed.). Pearson.
- Kotler, P., & Keller, K. L. (2016). Marketing Management (3rd ed.). Pearson.
- Siegel, E. (2016). Predictive Analytics: The Power to Predict Who Will Click, Buy, Lie, or Die. Wiley.
- Grigsby, M. (2022). Marketing Analytics: A Practical Guide to Improving Consumer Insights Using Data Techniques (3rd ed.). Kogan Page.
- Wedel, M., & Kannan, P. K. (2016). Marketing Analytics for Data-Rich Environments. Journal of Marketing (80) 6, 97-121.
- Lazer, D., & Radford, J. (2017). Data ex Machina: Introduction to Big Data. Annual Review of Sociology, 43, 19-39.
- Rich, W. (2025). Consumer Behavior Trends in 2025.
<https://www.globalbankingandfinance.com/consumer-behavior-trends-in-2025-a-comprehensive-analysis>.



AI-Driven Bioinformatics & Life Sciences

München

Modulbeschreibung

Precision Medicine

Modulnummer	Niveau Master	Studiengangsleitung Prof. Dr. Swati Chandna
ECTS (Benotung) 6 (benotet)	Dauer Semester	Häufigkeit 2 per Year
Sprache English	Workload Insgesamt: 150 Stunden - davon Kontaktzeit: 34 Stunden - davon Selbststudium: 116 Stunden - davon Praxis: 0 Stunden	Veranstaltungstyp Fallarbeit Gruppenarbeit Lehrvortrag Übung

Teilnahmevoraussetzungen

Kursinformation

Modulprüfung

Prüfungsteile	Gewichtung (Bewertung)
Projektarbeit	100%

Qualifikationsziele (Learning Outcomes) und Kompetenzen

Nach erfolgreichem Abschluss des Moduls können die Studierenden ...

Handlungskompetenz

This module introduces students to the principles, data types, and analytic methods used in precision medicine—an emerging field that integrates genomics, clinical data, and AI to provide patient-specific diagnostics and treatments. Students will gain hands-on experience in working with biomedical datasets, developing predictive models, and understanding ethical, legal, and social implications in personalized healthcare.

Fachkompetenz „Wissen und Verstehen“

- Understand the foundations of precision medicine, including genomic medicine, phenotyping, and stratified care.
- Differentiate between key biomedical data types: genomics, transcriptomics, EHRs, clinical biomarkers.
- Apply machine learning models (e.g., classification, clustering, survival analysis) to patient data for risk stratification and decision support.
- Interpret clinical relevance of models and results in the context of disease subtyping, treatment response, and prognosis prediction.
- Understand the role of data standards and interoperability (HL7, FHIR, OMOP CDM) in healthcare analytics.

Methodenkompetenz „Einsatz, Anwendung und Erzeugung von Wissen“

- Preprocess and analyze real-world biomedical datasets (e.g., TCGA, GEO, UK Biobank).
- Apply dimensionality reduction (PCA, t-SNE), clustering, and feature selection techniques for high-dimensional data.
- Build and evaluate predictive models using scikit-learn, lifelines, xgboost, or tensorflow/keras.
- Perform multi-omics integration and survival modeling where relevant.
- Apply fairness, explainability, and uncertainty estimation techniques to clinical ML models.

Selbstkompetenz „Wissenschaftliches Selbstverständnis und Professionalität“

- Reflect on the risks of bias, overfitting, and ethical implications when modeling healthcare data.
- Demonstrate autonomy in identifying a medical problem, choosing appropriate methods, and justifying analytical choices.
- Acknowledge uncertainty and limitations in clinical prediction and treatment recommendation settings.
- Maintain integrity when handling sensitive health data, adhering to data protection standards (e.g., GDPR, HIPAA).

Sozialkompetenz „Kommunikation und Kooperation“

- Collaborate in interdisciplinary teams combining clinical, bioinformatics, and data science expertise.
- Communicate patient-centered data insights in understandable language to both technical and non-technical stakeholders.
- Discuss sensitive data governance, patient consent, and ethical boundaries in medical AI applications.
- Reflect on the societal impact of personalized healthcare and equity in AI-driven medicine.

Constructive Alignment

- Lectures provide a theoretical foundation in genomics, patient data, and precision medicine strategy.
- Labs and guided exercises develop practical skills in data integration, prediction modeling, and medical ML evaluation.
- Group projects focus on solving a biomedical or healthcare personalization problem using real-world datasets.
- Discussions and reports support reflection on ethical, regulatory, and social issues in personalized healthcare.
- All activities and assessments promote competence development in clinical AI, health data science, and responsible decision-making in accordance with the CORE model.

Inhalt des Moduls

1. Introduction to Precision Medicine

- What is precision medicine? Historical evolution and use cases
- Disease heterogeneity and stratified treatment response
- Overview of patient-level data: omics, clinical, imaging, wearable

2. Data and Infrastructure in Precision Medicine

- Electronic health records (EHR) and clinical terminologies
- Genomic data formats (FASTA, VCF), high-throughput sequencing
- Interoperability: HL7, FHIR, OMOP, data warehouses in healthcare

3. Predictive Modeling for Individualized Care

- Supervised learning: diagnosis, treatment response prediction
- Unsupervised learning: patient clustering, phenotyping
- Survival analysis: Cox models, Kaplan-Meier, lifelines in Python
- Handling imbalanced, sparse, and high-dimensional data

4. Model Evaluation and Interpretation

- Metrics: AUROC, AUPRC, calibration curves
- SHAP, LIME, and interpretability in medical ML
- Model generalizability and reproducibility

5. Ethical, Legal, and Social Implications (ELSI)

- Informed consent, patient privacy, secondary data use
- Fairness and bias in clinical algorithms
- Regulation and compliance (e.g., EU AI Act, MDR, GDPR)
- Future of AI in personalized medicine and digital therapeutics

Literatur

- Schork, N. J. (2015). Personalized medicine: Time for one-person trials. *Nature*.
- Shickel, B. et al. (2018). Deep learning in electronic health records. *J*
- Hastie, T., Tibshirani, R., & Friedman, J. (2009). *The Elements of Statistical Learning*.
- Kermany, D. et al. (2018). Identifying medical diagnoses and treatable diseases by image-based deep learning.
- Topol, E. (2019). *Deep Medicine*.



Semester 2: Mandatory Modules

Heidelberg, Hamburg, München, Fürth

Modulbeschreibung

Data Management

Modulnummer	Niveau Master	Studiengangsleitung Prof. Dr. Swati Chandna
ECTS (Benotung) 6 (benotet)	Dauer Semester	Häufigkeit 2 per Year
Sprache English	Workload Insgesamt: 150 Stunden - davon Kontaktzeit: 48 Stunden - davon Selbststudium: 102 Stunden - davon Praxis: 0 Stunden	Veranstaltungstyp Gruppenarbeit Lehrvortrag Übung

Teilnahmevoraussetzungen

Kursinformation

Modulprüfung

Prüfungsteile	Gewichtung (Bewertung)
Projektarbeit	100%

Qualifikationsziele (Learning Outcomes) und Kompetenzen

Nach erfolgreichem Abschluss des Moduls können die Studierenden ...

Handlungskompetenz

This module equips students with the knowledge and tools to ensure that data used in analytics and AI is trustworthy, consistent, accessible, and structured appropriately. Emphasis is placed on data quality assurance, data cleaning workflows, metadata handling, graph data modeling, and the use of modern platforms such as Tableau Prep and Gephi for managing, visualizing, and communicating data integrity and structure.

Fachkompetenz „Wissen und Verstehen“

- Understand and apply principles of data quality, profiling, and cleaning
- Model and manage relational and graph-based data
- Use Tableau Prep, Gephi, and Python for profiling, cleaning, and visualizing datasets
- Differentiate between data lakes, data warehouses, and hybrid storage systems
- Recognize metadata standards and FAIR data management principles

Methodenkompetenz „Einsatz, Anwendung und Erzeugung von Wissen“

- Conduct detailed data profiling to assess quality and readiness
- Apply cleaning workflows using tools such as Tableau Prep and OpenRefine
- Create entity-relationship diagrams and graph models to organize structured and unstructured data
- Work with semi-structured formats (e.g., JSON, XML) in data lakes
- Visualize graph networks (e.g., relationships, influence, connectivity) with Gephi

Selbstkompetenz „Wissenschaftliches Selbstverständnis und Professionalität“

- Demonstrate professional responsibility in handling and correcting flawed data
- Reflect on the ethical and analytical risks of ignoring data quality issues
- Show initiative in identifying quality issues in real-world datasets

Sozialkompetenz „Kommunikation und Kooperation“

- Collaborate on data cleaning and documentation tasks
- Participate in peer evaluation of data quality improvement plans
- Communicate quality challenges and metadata needs to technical and business stakeholders

Constructive Alignment

- Lectures develop foundational knowledge in quality, modeling, and governance
- Hands-on labs with Tableau Prep, Open Refine, Gephi, and Python build technical proficiency
- Projects and presentations assess group collaboration and communication skills
- Ethical reflection tasks ensure awareness of responsible data stewardship
- All assessments are designed to demonstrate applied competence across real-world tools and methods

Inhalt des Moduls

1. Introduction to Data Management for Analytics

- Differences between data engineering and data quality roles
- Structured, semi-structured, and unstructured data
- The role of metadata and standards in data discovery and reuse
- Overview of tools: Tableau Prep, Gephi, OpenRefine, Python

2. Data Quality Dimensions and Metrics

- Completeness, accuracy, consistency, timeliness, uniqueness
- Creating quality rule sets and alerts
- KPIs for data quality (error rate, validation coverage, data freshness)

3. Data Profiling Techniques

- Column-level and dataset-level profiling
- Detection of outliers, duplicates, missing values, and inconsistencies
- Value distributions, frequency analysis, type mismatches

4. Data Cleaning Workflows

- Standardization and normalization
- Data type conversion and encoding
- Deduplication (exact and fuzzy matching)
- Imputation strategies: statistical, domain-based, ML-based

5. Data Modeling and Metadata

- Conceptual, logical, and physical data models
- Relational modeling: ERDs, normalization, schema design
- Introduction to metadata repositories and data catalogs
- Documenting lineage and versioning

6. Data Lakes, Warehouses, and Hybrid Architectures

- Data warehouse: star/snowflake schema, OLAP
- Data lake: schema-on-read, storage-first
- Lakehouse architecture: blending the two
- Use cases: historical reporting vs. real-time analytics

7. FAIR Principles, Ethics, and AI Readiness

- FAIR: Findable, Accessible, Interoperable, Reusable
- Ethics of data access and cleaning: bias, inference, consent
- Preparing data for trustworthy AI: transparency, traceability, reproducibility

Literatur

- Barton RD: Talend Open Studio Cookbook, Packt Publishing, 2013
- Blokdyk G: Data transformation: A Clear and Concise Reference, CreateSpace Independent Publishing Platform, 2018
- Cielen D & Meysman A: Introducing Data Science, Manning Verlag, 2016
- Halevy A et al.: Principles of Data Integration, Elsevier LTD, 2012.
- Mitchell MN: Data Management Using Stata: A Practical Handbook, Stata Press, 2010
- Verborgh R & De Wilde M: Using OpenRefine, Packt Publishing, 2013

- Recent research literature from peer-reviewed journals

Modulbeschreibung

Natural Language Processing

Modulnummer	Niveau Master	Studiengangsleitung Prof. Dr. Swati Chandna
ECTS (Benotung) 6 (benotet)	Dauer Semester	Häufigkeit 2 per Year
Sprache English	Workload Insgesamt: 150 Stunden - davon Kontaktzeit: 64 Stunden - davon Selbststudium: 86 Stunden - davon Praxis: 0 Stunden	Veranstaltungstyp Gruppenarbeit Lehrvortrag Seminar Übung

Teilnahmevoraussetzungen

Kursinformation

Modulprüfung

Prüfungsteile	Gewichtung (Bewertung)
Praktische Arbeit	100%

Qualifikationsziele (Learning Outcomes) und Kompetenzen

Nach erfolgreichem Abschluss des Moduls können die Studierenden ...

Handlungskompetenz

This module introduces students to the principles and techniques of Natural Language Processing (NLP)—a key subfield of AI focused on analyzing, understanding, and generating human language. Students will learn how to process and extract meaningful patterns from text data using a blend of linguistic rules, statistical models, and deep learning methods. The course combines foundational linguistics with practical machine learning, and prepares students to build NLP applications for domains such as customer feedback analysis, document classification, sentiment analysis, and conversational AI.

Fachkompetenz „Wissen und Verstehen“

- Understand linguistic concepts relevant to NLP: tokens, lemmas, syntax, semantics
- Describe core NLP tasks such as tokenization, POS tagging, NER, and sentiment analysis
- Explain and apply models such as TF-IDF, word embeddings, transformers, and fine-tuned LLMs
- Select appropriate NLP techniques for different application contexts (e.g., text classification, summarization)

Methodenkompetenz „Einsatz, Anwendung und Erzeugung von Wissen“

- Preprocess raw text for ML pipelines (cleaning, tokenization, vectorization)
- Use libraries like NLTK, spaCy, HuggingFace Transformers, and Scikit-learn
- Evaluate NLP models using precision, recall, F1, and BLEU/ROUGE for generation tasks
- Integrate NLP pipelines into applications (e.g., chatbots, dashboards, search engines)

Selbstkompetenz „Wissenschaftliches Selbstverständnis und Professionalität“

- Demonstrate responsible use of language data and awareness of language bias
- Critically assess model performance and interpretability
- Reflect on the ethical implications of language-based AI systems (e.g., hallucinations, stereotyping)

Sozialkompetenz „Kommunikation und Kooperation“

- Collaborate on real-world text analysis projects (e.g., media analysis, bias detection)
- Present NLP workflows and findings to diverse stakeholders
- Reflect on challenges and limitations when analyzing human language

Constructive Alignment

- Lectures and guided readings introduce students to key theoretical concepts in NLP, such as language modeling, embeddings, and neural architectures, supporting the development of specialist competence in understanding natural language structures and processing techniques.
- Hands-on lab sessions and coding exercises using real-world datasets (e.g., news articles, product reviews, social media content) enable students to apply preprocessing, classification, and information extraction workflows, building methodological competence.
- Use of Python libraries and NLP toolkits (e.g., spaCy, HuggingFace Transformers, Gensim) allows students to implement, fine-tune, and evaluate a range of NLP models, developing the technical depth necessary to handle diverse text analytics tasks.
- Collaborative mini-projects support social competence by encouraging peer-based learning, problem-solving, and the presentation of results to technical and non-technical audiences.
- Reflective reporting and ethical evaluation components foster personal competence by prompting students to consider the impact of NLP systems in real-world applications, including fairness, bias, and interpretability.
- Formative assessments (quizzes, code reviews) ensure ongoing feedback on student progress, while summative

assessments (e.g., final project with presentation and documentation) are designed to evaluate integrated learning outcomes across the CORE framework.

Inhalt des Moduls

- Refresher: Machine Learning from Analytics 1
- Python for Data Science
- Text Mining: Text Preprocessing, Feature Creation, Feature Selection, Pattern Discovery
- Natural Language Processing
- Text Processing
- Information Extraction
- String Similarity
- Information Retrieval
- Ranked Retrieval
- Model Selection
- Feature Engineering: Feature Reduction, Feature Scaling, Feature Encoding, Feature Selection

1. Introduction to NLP and Linguistic Foundations

- What is NLP? History, applications, and challenges
- Linguistic levels: phonetics, morphology, syntax, semantics, pragmatics
- Text as data: corpora, tokens, types, vocabulary
- Introduction to NLP tools: NLTK, spaCy, HuggingFace

2. Text Preprocessing and Representation

- Tokenization, lemmatization, stemming
- Stopword removal, regular expressions
- N-grams and bag-of-words
- Term weighting: TF, IDF, TF-IDF
- Vector space model and cosine similarity

3. Text Classification and Supervised Learning

- Feature engineering for text
- Naïve Bayes, logistic regression, SVM for document classification
- Sentiment analysis and opinion mining
- Hyperparameter tuning and pipeline automation

4. Named Entity Recognition (NER) and Information Extraction

- What is NER? Applications and challenges
- Rule-based vs. statistical and neural approaches
- Chunking, dependency parsing, relation extraction
- Intro to knowledge graphs (optional)

5. Topic Modeling and Unsupervised NLP

- Latent Dirichlet Allocation (LDA)
- Non-negative Matrix Factorization (NMF)
- Word frequency vs. latent structure
- Topic coherence and labeling

6. Word Embeddings and Vector Semantics

- Word2Vec, GloVe, FastText: skip-gram and CBOW
- Limitations of static embeddings

- Visualizing embeddings: t-SNE, PCA
- Measuring semantic similarity and analogies

7. Transformers and Pre-trained Language Models

- Self-attention and the Transformer architecture
- BERT, RoBERTa, DistilBERT: masked language models
- Fine-tuning and transfer learning
- Intro to prompt engineering (basic usage)

8. NLP for Text Generation and Summarization

- Text generation models (GPT, T5)
- Summarization: extractive vs. abstractive methods
- Evaluation metrics: BLEU, ROUGE
- Limitations: hallucination, repetitiveness, control

9. Ethical NLP and Bias Mitigation

- Bias in word embeddings and LLMs
- Offensive content and toxicity detection
- Transparency, fairness, and explainability in NLP models
- Ethical project design and deployment

Literatur

Core Texts

- Jurafsky, D., & Martin, J. H. (2023). Speech and Language Processing (3rd Ed. draft – <https://web.stanford.edu/~jurafsky/slp3/>)
- Bird, S., Klein, E., & Loper, E. (2009). Natural Language Processing with Python (NLTK Book)
- Goldberg, Y. (2017). Neural Network Methods in NLP
- Eisenstein, J. (2019). Introduction to Natural Language Processing

Supplementary & Practice

- Chowdhury, R. (2020). Practical Natural Language Processing
- HuggingFace Course – <https://huggingface.co/learn/nlp-course>
- fast.ai NLP Course – <https://course.fast.ai>

Modulbeschreibung

Responsible AI and Data Ethics

Modulnummer	Niveau Master	Studiengangsleitung Prof. Dr. Swati Chandna
ECTS (Benotung) 6 (benotet)	Dauer Semester	Häufigkeit 2 per Year
Sprache English	Workload Insgesamt: 150 Stunden - davon Kontaktzeit: 45 Stunden - davon Selbststudium: 105 Stunden - davon Praxis: 0 Stunden	Veranstaltungstyp Debatte Fallarbeit Gruppenarbeit Präsentation

Teilnahmevoraussetzungen

Kursinformation

Modulprüfung

Prüfungsteile	Gewichtung (Bewertung)
Projektarbeit	70%
Präsentation	30%

Qualifikationsziele (Learning Outcomes) und Kompetenzen

Nach erfolgreichem Abschluss des Moduls können die Studierenden ...

Handlungskompetenz

This module enables students to understand and address the ethical, legal, and societal implications of AI systems. They will critically engage with frameworks such as the EU AI Act, GDPR, and industry standards, and apply fairness, transparency, and privacy principles to real-world scenarios. Students will learn to identify and mitigate risks using tools for bias detection and privacy enhancement. Through case studies and group projects, they will evaluate responsible AI strategies and corporate governance models. The course also fosters interdisciplinary collaboration and ethical decision-making in diverse teams. By the end, students will be equipped to apply responsible AI practices in both technical and organizational contexts.

Fachkompetenz „Wissen und Verstehen“

- Students will develop the ability to critically analyze the development, deployment, and societal impact of AI systems from an ethical perspective, grounded in normative frameworks such as consequentialism, deontology, and virtue ethics.
- They will learn to apply ethical reasoning and privacy principles to complex socio-technical scenarios, including automated decision-making, surveillance, and algorithmic bias.
- The module provides in-depth knowledge of key AI-related laws and standards, including the EU AI Act, GDPR, OECD Principles, and IEEE Ethically Aligned Design, enabling students to evaluate compliance requirements and regulatory implications for AI projects.
- Students will understand the fundamentals of data governance and responsible data use, including data minimization, consent management, and the ethical curation of training data.
- They will gain hands-on experience with tools and techniques for detecting and mitigating algorithmic bias (e.g., Fairlearn, IBM AI Fairness 360) and for implementing privacy-preserving AI using technologies such as differential privacy, federated learning, and homomorphic encryption.
- The module introduces Explainable AI (XAI) methods such as SHAP and LIME, enabling students to enhance transparency and accountability in machine learning models.
- Through case studies and applied projects, students will be equipped to identify ethical risks in real-world AI applications across domains and to design responsible AI solutions that align with legal, organizational, and societal expectations.

Methodenkompetenz „Einsatz, Anwendung und Erzeugung von Wissen“

- Apply fairness, privacy, and bias detection tools (e.g., Aequitas, IBM AI Fairness 360, OpenDP)
- Conduct ethical impact assessments for AI applications
- Perform qualitative and quantitative risk analysis of AI systems
- Communicate the ethical dimensions of AI projects using structured frameworks

Selbstkompetenz „Wissenschaftliches Selbstverständnis und Professionalität“

- Reflect on one's role and responsibility in the AI lifecycle
- Evaluate the tension between innovation and ethical constraints
- Demonstrate integrity, transparency, and accountability in technology design and use
- Acknowledge personal values and biases in ethical decision-making

Sozialkompetenz „Kommunikation und Kooperation“

- Collaborate on team-based case analyses and ethical evaluations
- Engage in multi-perspective debates on controversial AI applications

-Translate ethical and legal considerations into actionable strategies for technical and managerial audiences

Constructive Alignment

-Lectures and case studies establish theoretical, legal, and practical knowledge about ethical and responsible AI development (Specialist Competence).

-Hands-on labs with open-source fairness and privacy tools reinforce analytical and methodological skills in real-world data contexts (Methodological Competence).

-Group projects foster collaboration and communication skills through interdisciplinary discussion, stakeholder mapping, and collective problem-solving (Social Competence).

-Reflective writing tasks help students consider the ethical implications of their personal choices and roles in AI development (Personal Competence).

-Assessment tasks require students to analyze current AI applications, identify ethical risks, and propose implementable mitigation strategies, directly aligned with CORE learning outcomes.

Inhalt des Moduls

1. Introduction to Responsible AI and Data Ethics

- Core ethical theories: consequentialism, deontology, virtue ethics
- Understanding ethical dilemmas in AI systems
- Societal impacts: autonomy, justice, human rights, and public trust
- Key ethical frameworks and codices

2. Current Developments in AI Legislation

- Deep dive: EU AI Act
- Risk-based classification of AI systems
- Obligations for providers and users
- GDPR: relevance to AI
- Automated decision-making, right to explanation, data minimization
- Global overview: US, China, UNESCO, and national AI strategies
- Legal responsibilities of AI developers, deployers, and controllers

3. Standards and Norms in AI Governance

- Introduction to AI standardization
- Role of CEN-CENELEC, IEEE, NIST, and industry consortia
- Implications of standard adoption on AI development pipelines

4. Responsible AI in Industry Practice

- Organizational strategies for ethical AI:
- Principles, policies, audit frameworks
- Ethics boards and transparency reports
- Comparative review: Google, Microsoft, IBM, SAP, and others
- Case studies: Facial recognition, automated hiring, content moderation failures

5. Privacy and Security in AI Systems

- Introduction to privacy-enhancing technologies (PETs):
- Data anonymization, pseudonymization, and tokenization
- Secure AI lifecycle: threats, threat modeling, and privacy-by-design
- Trade-offs between data utility and privacy protection

6. Bias, Fairness, and Transparency in AI

- Understanding sources and types of bias: data, algorithmic, societal
- Fairness criteria: demographic parity, equal opportunity, individual fairness
- Bias detection and mitigation tools
- Explainability and transparency:

7. Ethical Challenges: Misuse of AI and Data: Case studies

- Cambridge Analytica
- AI surveillance systems in public security
- Deepfakes and manipulation in political contexts
- Risks of opaque automation: credit scoring, predictive policing
- Ethical issues in media consumption: echo chambers, filter bubbles

-Balancing innovation and societal safeguards

8. AI Risk Management and Mitigation

- Identifying ethical and societal risks in AI systems
- Ethical risk analysis frameworks: probability, severity, stakeholder impact
- Risk mitigation strategies:
 - Technical: validation, robustness, human-in-the-loop
 - Organizational: audits, documentation, impact assessments

Literatur

- Floridi, L. (2019). The Ethics of Artificial Intelligence. Oxford University Press.
- Mittelstadt, B. & Floridi, L. (2016). The Ethics of Algorithms: Mapping the Debate. Big Data & Society.
- Russell, S. (2019). Human Compatible: Artificial Intelligence and the Problem of Control. Viking.
- European Commission (2021). Proposal for a Regulation Laying Down Harmonized Rules on Artificial Intelligence (EU AI Act).
- European Data Protection Board (2020). Guidelines on Automated Decision-Making and Profiling Under GDPR.
- O'Neil, C. (2016). Weapons of Math Destruction: How Big Data Increases Inequality and Threatens Democracy. Crown Publishing.
- Molnar, C. (2022) Interpretable Machine Learning: A Guide for Making Black Box Models Explainable (2nd ed.).
- Jobin, A., Ienca, M., & Vayena, E. (2019). The Global Landscape of AI Ethics Guidelines. Nature Machine Intelligence.
- Madaio, M. A., et al. (2020). Co-Designing Checklists to Understand Organizational Challenges and Opportunities around Fairness in AI. -CHI Conference on Human Factors in Computing Systems.
- Schwartmann/Keber/Zenner (2024) KI-VO- Leitfaden für die Praxis (2., aktualisierte Auflage)
- Spiekermann(2023)-Value-Based- Engineering
- Diverse DIN & ISO-Standards



Semester 2: General or Interdisciplinary Track

Campus Specific



General Track

Heidelberg

Modulbeschreibung

Collaborative Industry Project - I

Modulnummer	Niveau Master	Studiengangsleitung Prof. Dr. Swati Chandna
ECTS (Benotung) 6 (benotet)	Dauer Semester	Häufigkeit 2 per Year
Sprache English	Workload Insgesamt: 150 Stunden - davon Kontaktzeit: 34 Stunden - davon Selbststudium: 116 Stunden - davon Praxis: 0 Stunden	Veranstaltungstyp Fallarbeit Fallbearbeitung und Reflexion Gruppenarbeit Projektentwicklung Übung

Teilnahmevoraussetzungen

Kursinformation

Modulprüfung

Prüfungsteile	Gewichtung (Bewertung)
Projektarbeit	100%

Qualifikationsziele (Learning Outcomes) und Kompetenzen

Nach erfolgreichem Abschluss des Moduls können die Studierenden ...

Handlungskompetenz

This module provides students with hands-on experience in solving real-world data analytics problems in collaboration with industry partners. Students apply data science methods—including data cleaning, exploration, modeling, and visualization—to authentic datasets provided by businesses or organizations. The project-based format fosters critical thinking, problem-solving, and communication skills while exposing students to stakeholder engagement and industry-relevant workflows. To support project success and professional development, students participate in targeted training sessions on SCRUM project management, conflict resolution, and presentation skills, equipping them with the interpersonal and organizational capabilities essential for team-based data work.

By the end of the module, students will be able to:

- Apply analytical techniques to solve a practical industry data problem.
- Collaborate in interdisciplinary teams and manage project timelines.
- Communicate data-driven insights effectively to technical and non-technical stakeholders.
- Understand the business value of data analytics in decision-making contexts.
- Reflect on the ethical, organizational, and technical dimensions of real-world data work.

Fachkompetenz „Wissen und Verstehen“

- Understand business problems and translate them into data analysis objectives.
- Select and apply suitable data analytics methods (e.g., regression, clustering, forecasting, anomaly detection).
- Interpret and communicate analytical results in an industry-relevant context.
- Work with structured, semi-structured, or time-series data from domains such as retail, logistics, healthcare, or finance.

Methodenkompetenz „Einsatz, Anwendung und Erzeugung von Wissen“

- Manage project workflows using agile practices or CRISP-DM.
- Design reproducible and modular data workflows using tools like Python, Jupyter, or Power BI.
- Evaluate model performance based on business metrics and constraints.
- Document project work clearly and professionally.

Selbstkompetenz „Wissenschaftliches Selbstverständnis und Professionalität“

- Take initiative in problem-solving, planning, and resource allocation.
- Reflect on individual contributions and team dynamics.
- Demonstrate professional responsibility and ethical awareness in handling business data.

Sozialkompetenz „Kommunikation und Kooperation“

- Collaborate in teams using shared version control and task management platforms (e.g., GitHub, Trello).
- Communicate with external stakeholders (e.g., company mentors or clients).
- Provide constructive peer feedback and coordinate responsibilities.

Constructive Alignment

- Technical workshops and SCRUM training reinforce methodological competence and structure project workflows.
- Case-based learning with client feedback helps develop specialist insight and practical relevance.
- Conflict resolution and feedback sessions foster social competence and communication skills.
- Final presentations and reflection reports encourage personal accountability and growth.

Inhalt des Moduls

1. Project Kick-Off & Team Formation

- Overview of project goals, expectations, and assessment
- Introduction to client or simulated case
- Team composition and skills inventory
- Training session: SCRUM for Data Projects
- Roles (Scrum Master, Product Owner, Team Members)
- Sprint planning and backlog management tools (e.g., Trello, GitHub Projects)

2. Business Understanding & Communication

- Stakeholder analysis and requirement clarification
- Defining KPIs and success criteria
- Communication protocols in team and with clients
- Training session: Professional Presentation & Reporting Skills

3. Data Understanding and Processing

- Data access, privacy handling, and quality checks
- Exploratory data analysis and feature engineering
- Data visualization for understanding trends and outliers

4. Modeling and Evaluation

- Choice of models (e.g., regression, clustering, forecasting, classification)
- Evaluation strategies: performance metrics, stakeholder relevance
- Communicating model uncertainty and business impact

5. Conflict Resolution & Feedback Integration

- Training session: Conflict Management for Project Teams
- Identifying and de-escalating conflict
- Feedback culture and communication techniques
- Mid-project retrospectives and team feedback loops

6. Final Delivery & Reflection

- Preparation of project deliverables: technical report, dashboard/app, executive summary
- Client or panel presentation (pitch or demo)
- Peer and self-assessment
- Written reflection on learning outcomes and team experience

Literatur

- Provost, F., & Fawcett, T. (2013). Data Science for Business: What You Need to Know about Data Mining and Data-Analytic Thinking
- Marr, B. (2016). Big Data in Practice: How 45 Successful Companies Used Big Data Analytics to Deliver Extraordinary Results
- Shmueli, G., Bruce, P. C., & Patel, N. R. (2017). Data Mining for Business Analytics: Concepts, Techniques, and Applications in Python
- Sutherland, J. (2014). Scrum: The Art of Doing Twice the Work in Half the Time
- Cockburn, A. (2007). Agile Software Development: The Cooperative Game

Modulbeschreibung

Collaborative Industry Project - II

Modulnummer	Niveau Master	Studiengangsleitung Prof. Dr. Swati Chandna
ECTS (Benotung) 6 (benotet)	Dauer Semester	Häufigkeit 2 per Year
Sprache English	Workload Insgesamt: 150 Stunden - davon Kontaktzeit: 34 Stunden - davon Selbststudium: 116 Stunden - davon Praxis: 0 Stunden	Veranstaltungstyp Fallarbeit Fallbearbeitung und Reflexion Gruppenarbeit Projektentwicklung Übung

Teilnahmevoraussetzungen

Kursinformation

Modulprüfung

Prüfungsteile	Gewichtung (Bewertung)
Projektarbeit	100%

Qualifikationsziele (Learning Outcomes) und Kompetenzen

Nach erfolgreichem Abschluss des Moduls können die Studierenden ...

Handlungskompetenz

This module advances students from exploratory data analysis and problem scoping (covered in Project I) to the development and deployment of AI-based solutions in response to real-world industry or institutional challenges. Students will implement full-stack machine learning or AI pipelines, incorporate fairness and privacy considerations, and present a working prototype to stakeholders. Emphasis is placed on technical robustness, ethical deployment, operational feasibility, and project ownership.

Fachkompetenz „Wissen und Verstehen“

- Design end-to-end AI solutions from structured/unstructured data using appropriate ML or deep learning models
- Implement fairness, privacy, and explainability into model pipelines
- Evaluate deployment feasibility and integrate AI components into functional systems
- Work with advanced tools such as model registries, APIs, or low-code deployment platforms

Methodenkompetenz „Einsatz, Anwendung und Erzeugung von Wissen“

- Apply model versioning, testing, and evaluation practices (e.g., MLflow, Git, CI/CD)
- Build scalable and modular AI pipelines with reproducibility and monitoring in mind
- Integrate responsible AI workflows (e.g., bias detection, model explainability, privacy-preserving training)
- Use prototyping tools (e.g., Streamlit, FastAPI, Docker) for deploying AI services

Selbstkompetenz „Wissenschaftliches Selbstverständnis und Professionalität“

- Demonstrate ownership of specific AI components in the final solution
- Reflect on technical limitations, ethical trade-offs, and stakeholder alignment
- Uphold transparency and accountability in building and documenting AI systems
- Communicate complex AI decisions responsibly and accessibly

Sozialkompetenz „Kommunikation und Kooperation“

- Take on defined team roles (e.g., AI engineer, model validator, deployment lead) to ensure clear accountability
- Lead stakeholder engagement meetings and translate requirements into design choices
- Facilitate team feedback sessions and sprint retrospectives with reflection on conflict resolution
- Guide ethical discussions and assess real-world impact with collaborators

Constructive Alignment

- Workshops provide hands-on support for technical implementation, MLOps, and deployment
- Team sprints and retrospectives support role-based learning and project accountability
- Stakeholder demo days and pitch sessions assess application relevance and presentation competence
- Ethical reflection logs ensure students connect responsible AI theory with practical decisions

Inhalt des Moduls

1. Project Framing and Responsible AI Planning

- Technical problem scoping and stakeholder alignment
- Success criteria and risk identification
- Legal and ethical considerations (GDPR, EU AI Act, ISO/IEC alignment)
- Planning for explainability, fairness, and data stewardship

2. End-to-End AI Model Development

- Advanced model design (regression, classification, NLP, deep learning)
- Pipeline modularization and testing
- Incorporating fairness, bias mitigation, and model explainability tools (SHAP, LIME)
- Version control and experiment tracking (e.g. MLflow)

3. Dashboards and Visual Communication of AI Insights

- Principles of effective data storytelling and decision-focused dashboards
- Tools: Power BI, Tableau, Streamlit, Plotly Dash
- Visualizing model results, performance metrics, and fairness indicators
- Real-time model monitoring dashboards and feedback integration
- Designing interactive visual elements for technical and business stakeholders

4. Deployment and MLOps Foundations

- CI/CD integration, containerization (Docker), and cloud deployment workflows
- Hosting models and dashboards via web apps or internal servers
- Logging, performance monitoring, and maintaining model explainability post-deployment
- Governance of deployed models: updates, rollback, compliance

5. Responsible AI in Deployment Contexts

- Incorporate transparency, privacy, and fairness into final solution
- Document and justify trade-offs (e.g. accuracy vs interpretability)
- Validate solutions against ethical and legal compliance frameworks (ALTAI, ISO/IEC 42001)

6. Final Delivery & Stakeholder Communication

- Live demonstrations of working models and dashboards
- Business impact summary and return-on-investment narrative
- Stakeholder engagement through interactive dashboards and guided walkthroughs
- Ethical impact reflection and lessons learned
- Team presentation, peer review, and project debrief

Literatur

- Chip Huyen (2022). Designing Machine Learning Systems
- Mark Treveil & S. Razavi (2020). Introducing MLOps
- O'Reilly (2020). Building Machine Learning Powered Applications – E. Allam
- Mittelstadt, B. (2019). Principles Alone Cannot Guarantee Ethical AI – Nature Machine Intelligence



Creative AI and Media Analytics

Heidelberg

Modulbeschreibung

Social Media Analytics and Sentiment Analysis

Modulnummer	Niveau Master	Studiengangsleitung Prof. Dr. Swati Chandna
ECTS (Benotung) 6 (benotet)	Dauer Semester	Häufigkeit 2 per Year
Sprache English	Workload Insgesamt: 150 Stunden - davon Kontaktzeit: 34 Stunden - davon Selbststudium: 116 Stunden - davon Praxis: 0 Stunden	Veranstaltungstyp Gruppenarbeit Lehrvortrag

Teilnahmevoraussetzungen

Kursinformation

Modulprüfung

Prüfungsteile	Gewichtung (Bewertung)
Projektarbeit	100%

Qualifikationsziele (Learning Outcomes) und Kompetenzen

Nach erfolgreichem Abschluss des Moduls können die Studierenden ...

Handlungskompetenz

Students will gain a robust understanding of social media analytics and sentiment analysis, mastering both theoretical concepts and practical skills with cutting-edge tools. The goal is to enable learners to collect, analyze, and visualize social media data, deriving actionable insights into trends, sentiments, and behaviors. Students will critically interpret findings, address ethical challenges like bias and misinformation, and apply their skills to real-world contexts such as brand perception and public opinion. Through hands-on projects, they will develop adaptability and a responsible, data-driven approach, preparing them for future roles in data science and media intelligence.

Fachkompetenz „Wissen und Verstehen“

Students will develop advanced expertise in collecting, cleaning, and analyzing social media data using cutting-edge techniques such as text mining, natural language processing (NLP), and sentiment analysis (rule-based, machine learning, and deep learning approaches). They will gain proficiency in leveraging platform-specific APIs, designing efficient data pipelines, and applying statistical and computational methods to evaluate public opinion, brand perception, and emerging trends with precision. Additionally, students will learn to identify and mitigate biases, detect bots and misinformation, and integrate interdisciplinary insights from psychology and communication studies to enrich their analyses. This competence equips them to deliver high-quality, actionable outcomes in professional settings such as market research, digital strategy, and societal impact assessment.

Methodenkompetenz „Einsatz, Anwendung und Erzeugung von Wissen“

Students will develop and apply advanced text analysis, machine learning classifiers, and sentiment models using Python tools. They will build efficient data workflows, create real-time sentiment dashboards, and innovate with methods like ensemble modeling. Adaptable to evolving data challenges, they will produce robust, professional-grade results..

Selbstkompetenz „Wissenschaftliches Selbstverständnis und Professionalität“

Students build resilience in dealing with noisy and biased social media data. They independently reflect on ethical challenges and limitations in sentiment interpretation

Sozialkompetenz „Kommunikation und Kooperation“

Students work in diverse teams to analyze multi-perspective sentiment data and share insights through presentations and reports. Collaboration and peer learning are emphasized.

Constructive Alignment

Students engage in hands-on project work using real-world social media datasets. The project includes data collection, model development, and presentation of results. The module strengthens digital skills, critical reflection, and teamwork through iterative feedback and peer collaboration

Inhalt des Moduls

- Introduction to Social Media Analytics
- Social Media Platforms and Data Types
- Data Collection: APIs, Scraping, and Ethics
- Text Preprocessing and Cleaning
- Natural Language Processing (NLP) for Social Media
- Sentiment Analysis Techniques (Rule-based, ML, DL)
- Emotion Detection and Opinion Mining
- Hashtag and Trend Analysis
- Topic Modeling (LDA, BERTopic)
- Network Analysis and Influence Detection
- Bot Detection and Fake News Identification
- Visual Analytics and Dashboards
- Real-Time Social Media Monitoring
- Bias, Privacy, and Ethics in SMA
- Case Studies and Practical Applications
- Hands-on Projects with Real Social Media Data

Literatur

- Liu, B. (2015). Sentiment Analysis: Mining Opinions, Sentiments, and Emotions. Cambridge University Press.
- Russell, M. A. (2013). Mining the Social Web. O'Reilly Media.
- Pak, A., & Paroubek, P. (2010). Twitter as a Corpus for Sentiment Analysis.
- Bird, S., Klein, E., & Loper, E. (2009). Natural Language Processing with Python. O'Reilly Media.
- McKinney, W. (2017). Python for Data Analysis. O'Reilly Media

Modulbeschreibung

Generative AI for Creative Content

Modulnummer	Niveau Master	Studiengangsleitung Prof. Dr. Swati Chandna
ECTS (Benotung) 6 (benotet)	Dauer Semester	Häufigkeit 2 per Year
Sprache English	Workload Insgesamt: 150 Stunden - davon Kontaktzeit: 34 Stunden - davon Selbststudium: 116 Stunden - davon Praxis: 0 Stunden	Veranstaltungstyp Diskussion Gruppenarbeit Seminar

Teilnahmevoraussetzungen

Kursinformation

Modulprüfung

Prüfungsteile	Gewichtung (Bewertung)
Projektarbeit	70%
Präsentation	30%

Qualifikationsziele (Learning Outcomes) und Kompetenzen

Nach erfolgreichem Abschluss des Moduls können die Studierenden ...

Handlungskompetenz

This module explores the use of generative artificial intelligence (GenAI) models in creating digital content across modalities such as text, images, music, video, and design. Students learn how generative models work, how to fine-tune and prompt them, and how to evaluate the artistic, ethical, and social implications of their use. The course combines theoretical understanding of models like GANs, VAEs, and transformers with hands-on projects using state-of-the-art tools in creative domains.

Fachkompetenz „Wissen und Verstehen“

- Study and implement various generative models such as GANs, VAEs, and diffusion models.
- Explore multimodal generative AI applications in text, image, video, and audio synthesis
- Analyze model architectures, loss functions, and convergence behaviors.
- Gain proficiency in frameworks like TensorFlow, PyTorch, or JAX.
- Leverage GPU/TPU acceleration, distributed training, and model parallelism techniques.
- Understand underlying probability theories, variational inference, and latent space representations.
- Integrate cutting-edge research papers and emerging methods into practical projects.

Methodenkompetenz „Einsatz, Anwendung und Erzeugung von Wissen“

- Apply pre-trained GenAI tools such as ChatGPT, Midjourney, DALL-E, RunwayML, and Stable Diffusion
- Fine-tune models and design prompt engineering strategies
- Combine generative models with rule-based or traditional creative workflows
- Use evaluation techniques for generative output (BLEU, FID, human-in-the-loop)

Selbstkompetenz „Wissenschaftliches Selbstverständnis und Professionalität“

- Reflect on originality, ownership, and value in AI-generated content
- Evaluate the ethical use of GenAI in media, including concerns of bias, misinformation, and deepfakes
- Develop a critical lens on creativity, human-AI co-creation, and automation of artistic labor

Sozialkompetenz „Kommunikation und Kooperation“

- Collaborate on interdisciplinary creative projects with shared ownership
- Present generative works and discuss artistic choices and technical design
- Consider audience perception, cultural context, and collaborative authorship

Constructive Alignment

- Lectures and demonstrations provide theoretical and technical grounding in generative models (Specialist)
- Hands-on labs and guided tool use develop students' ability to work with GenAI tools for content generation (Methodological)
- Group projects promote collaborative creative processes and interdisciplinary teamwork (Social)
- Reflective essays and critique sessions foster ethical, aesthetic, and philosophical inquiry into AI-mediated creativity (Personal)

Inhalt des Moduls

1. Introduction to Generative AI

- What is Generative AI? Historical overview and modern developments
- Creative domains impacted by GenAI (text, image, video, audio, design)
- AI vs. human creativity: augmentation, collaboration, automation

2. Generative Models and Architectures

- GANs (Generative Adversarial Networks): DCGAN, StyleGAN
- Variational Autoencoders (VAEs) for latent space modeling
- Transformer-based models (GPT, T5, BERT, MusicLM)
- Diffusion models and their applications

3. Text Generation and Prompt Engineering

- Text generation with GPT-3/4, LLaMA, Claude
- Prompt design patterns for creativity, dialogue, storytelling
- Evaluation of generated text (BLEU, perplexity, content diversity)

4. Image Generation and Editing

- DALL-E 2, Midjourney, Stable Diffusion: model comparison
- Text-to-image generation: compositionality, aesthetics, realism

5. Audio and Music Synthesis

- AI-generated music: MusicLM, Jukebox, Riffusion
- Voice synthesis and style transfer

6. Video and Animation with GenAI

- Video diffusion, motion capture, frame prediction

7. Multimodal Creativity and Interactive Experiences

- Text-to-music, text-to-video pipelines
- Generative storytelling experiences and AI companions

8. Evaluation and Critique of Creative AI Outputs

- Quantitative and qualitative assessment: FID, human feedback, narrative coherence
- Creative Turing tests and public reception

Literatur

- Goodfellow, I., Pouget-Abadie, J., Mirza, M., Xu, B., Warde-Farley, D., Ozair, S., Courville, A., & Bengio, Y. (2014). Generative adversarial nets. *Advances in Neural Information Processing Systems*, 27.
- Kingma, D. P., & Welling, M. (2013). Auto-Encoding Variational Bayes. *arXiv preprint arXiv:1312.6114*.
- Radford, A., Metz, L., & Chintala, S. (2015). Unsupervised representation learning with deep convolutional generative adversarial networks. *arXiv preprint arXiv:1511.06434*.
- Salimans, T., Goodfellow, I., Zaremba, W., Cheung, V., Radford, A., & Chen, X. (2016). Improved techniques for training GANs. *arXiv preprint arXiv:1606.03498*.
- Ho, J., Jain, A., & Abbeel, P. (2020). Denoising diffusion probabilistic models. *arXiv preprint arXiv:2006.11239*.
- Karras, T., Laine, S., & Aila, T. (2019). A style-based generator architecture for generative adversarial networks. In

Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (pp. 4401-4410).

- Brock, A., Donahue, J., & Simonyan, K. (2018). Large scale GAN training for high fidelity natural image synthesis. In International Conference on Learning Representations (ICLR).



Supply Chain and Logistics Analytics

Hamburg

Modulbeschreibung

Inventory Analytics

Modulnummer	Niveau Master	Studiengangsleitung Prof. Dr. Swati Chandna
ECTS (Benotung) 6 (benotet)	Dauer Semester	Häufigkeit 2 per Year
Sprache English	Workload Insgesamt: 150 Stunden - davon Kontaktzeit: 34 Stunden - davon Selbststudium: 116 Stunden - davon Praxis: 0 Stunden	Veranstaltungstyp Lehrvortrag Übung

Teilnahmevoraussetzungen

Kursinformation

Modulprüfung

Prüfungsteile	Gewichtung (Bewertung)
Projektarbeit	70%
Präsentation	30%

Qualifikationsziele (Learning Outcomes) und Kompetenzen

Nach erfolgreichem Abschluss des Moduls können die Studierenden ...

Handlungskompetenz

- Understand inventory decision-making frameworks through a data science lens
- Translate inventory-related business problems into data analysis tasks
- Select and apply forecasting, clustering, and classification methods to inventory datasets
- Analyze inventory dynamics (turnover, aging, variability) to inform stocking policies

Fachkompetenz „Wissen und Verstehen“

- Understand core concepts such as EOQ, safety stock, reorder points, service levels, and inventory turnover
- Apply inventory classification (ABC/XYZ), demand variability, and lead time analytics
- Evaluate and select appropriate inventory policies for single and multi-echelon systems
- Interpret the impact of forecasting, seasonality, and demand uncertainty on inventory planning

Methodenkompetenz „Einsatz, Anwendung und Erzeugung von Wissen“

- Build time-series and machine learning models (ARIMA, Prophet, XGBoost, k-Means) for inventory prediction
- Use Python libraries for exploratory analysis, simulation, and dashboard development
- Evaluate model performance with metrics relevant to stockout risk, service level, and demand variability
- Design interpretable and robust models for real-world inventory environments

Selbstkompetenz „Wissenschaftliches Selbstverständnis und Professionalität“

- Reflect on the role of data science in driving efficient, ethical, and sustainable inventory systems
- Take initiative in model validation, uncertainty assessment, and scenario planning
- Demonstrate awareness of limitations and bias in demand data or stock movement logs

Sozialkompetenz „Kommunikation und Kooperation“

- Collaborate in small teams to analyze business inventory challenges
- Communicate complex modeling outcomes to logistics and operations managers
- Discuss the trade-offs between model accuracy, explainability, and operational feasibility

Constructive Alignment

- Lectures and real-data exercises build specialist competence in analyzing and interpreting inventory-related data
- Hands-on coding labs and modeling projects develop methodological fluency in forecasting, clustering, and simulation
- Team-based project and dashboard presentation cultivate communication, applied storytelling, and business translation (social competence)
- Reflection and ethical discussions foster personal growth in sustainable, fair, and responsible inventory planning decisions

Inhalt des Moduls

1. Data-Driven Inventory Management: Concepts & Objectives

- Inventory classification (A/B/C, variability-based)
- Operational KPIs: fill rate, turnover, cycle time
- Inventory as a system: demand, lead time, policies, and cost dynamics

2. Inventory Data Sources and Wrangling

- Raw ERP/warehouse data formats (SKUs, timestamps, quantities)
- Data merging: sales, stock movement, forecast vs. actual
- Feature engineering: moving averages, lead time distributions, demand profiles

3. Time-Series Forecasting for Inventory

- Baseline and classical methods: Moving Average, Holt-Winters
- ARIMA/SARIMA models for structured forecasting
- Facebook Prophet and NeuralProphet for scalable forecasting
- Forecast accuracy and error impact on inventory levels

4. Machine Learning for Inventory Decision Support

- Supervised learning for reorder prediction: logistic regression, decision trees, XGBoost
- Predicting safety stock needs using regression models
- Inventory churn prediction
- Interpreting models using SHAP/LIME

5. Clustering and Segmentation in Inventory Analytics

- Unsupervised learning: k-Means, DBSCAN for SKU grouping
- Segmentation by demand variability, stock age, sales pattern
- Data-driven inventory policy assignment via clustering

6. Simulation and Scenario Analysis

- Monte Carlo simulation of reorder points under uncertainty
- Modeling lead time variation and stockout impact
- What-if analysis for safety stock and service level optimization

7. Inventory Optimization with Data Constraints

- Linear and mixed-integer programming for replenishment decisions
- Cost minimization under capacity/service level constraints
- Solver libraries: PuLP (Python), Gurobi (optional introduction)

8. Visualization and Decision Dashboards

- Designing KPI dashboards for inventory health
- Visualizing uncertainty, reorder points, and aging stock

Literatur

- Vandeput, N. (2021). Data Science for Supply Chain Forecasting. Berlin, Boston: De Gruyter.
<https://doi.org/10.1515/9783110671124>
- Rossi, Roberto. (2021). Inventory Analytics. 10.11647/obp.0252.
- Silver, E. A. et al. (2016). Inventory and Production Management in Supply Chains

- Fildes, R. & Goodwin, P. (2007). Against Your Better Judgment? How Organizations Can Improve Their Use of Management Forecasts
- Chopra, S. (2019). Supply Chain Management: Strategy, Planning, and Operation

Modulbeschreibung

Logistics and Transportation Analytics

Modulnummer	Niveau Master	Studiengangsleitung Prof. Dr. Swati Chandna
ECTS (Benotung) 6 (benotet)	Dauer Semester	Häufigkeit 2 per Year
Sprache English	Workload Insgesamt: 150 Stunden - davon Kontaktzeit: 34 Stunden - davon Selbststudium: 116 Stunden - davon Praxis: 0 Stunden	Veranstaltungstyp Diskussion Gruppenarbeit

Teilnahmevoraussetzungen

Kursinformation

Modulprüfung

Prüfungsteile	Gewichtung (Bewertung)
Projektarbeit	100%

Qualifikationsziele (Learning Outcomes) und Kompetenzen

Nach erfolgreichem Abschluss des Moduls können die Studierenden ...

Handlungskompetenz

This module equips students with the skills to apply data analytics and optimization techniques to logistics and transportation challenges. Emphasis is placed on using real-world logistics datasets, network analysis, routing algorithms, and geospatial tools to improve transportation efficiency, reduce costs, and support sustainable operations. Students will gain practical experience in vehicle routing, freight modeling, delivery scheduling, and multimodal logistics using Python, open-source tools, and visual analytics platforms.

Fachkompetenz „Wissen und Verstehen“

- Understand the structure and components of logistics networks (e.g., hubs, routes, modes, and flow types)
- Analyze real-world transport operations using descriptive, predictive, and prescriptive analytics
- Model routing, fleet scheduling, and logistics flows using optimization and simulation
- Evaluate key logistics performance indicators (KPI): cost, delivery time, fuel use, service level

Methodenkompetenz „Einsatz, Anwendung und Erzeugung von Wissen“

- Apply machine learning and optimization models to routing and transport scheduling problems
- Work with tools such as Python (e.g., OR-Tools, NetworkX, Geopandas), QGIS, and Power BI
- Solve classical and modern transportation problems (e.g., TSP, VRP, last-mile delivery)
- Use GIS and geospatial data for mapping logistics operations

Selbstkompetenz „Wissenschaftliches Selbstverständnis und Professionalität“

- Reflect on trade-offs between efficiency, sustainability, and equity in transport decisions
- Demonstrate initiative in solving unstructured logistics problems using data
- Consider ethical and environmental factors in route planning and modal selection

Sozialkompetenz „Kommunikation und Kooperation“

- Collaborate in interdisciplinary teams on logistics case studies and simulations
- Present analytical findings and strategies to operations stakeholders
- Communicate technical constraints and policy trade-offs clearly in logistics contexts

Constructive Alignment

- Lectures and tool demos provide theory and technical knowledge in logistics analytics and network modeling (Specialist)
- Coding labs and GIS mapping exercises strengthen quantitative and spatial problem-solving skills (Methodological)
- Group projects and case studies build collaboration and communication skills for operations decision-making (Social)
- Ethical reflections and trade-off discussions support critical evaluation of sustainability and access issues (Personal)

Inhalt des Moduls

1. Introduction to Logistics and Transportation Systems

- Role of transportation in supply chains
- Modes of transport and logistics infrastructure
- Strategic vs. tactical vs. operational logistics decisions
- Emerging trends: urban logistics, decarbonization, digitization

2. Logistics KPIs and Data Analytics Foundations

- Delivery time, cost, fuel consumption, reliability, carbon footprint
- Data sources: IoT, telematics, ERP, GPS, open datasets
- Exploratory analysis of logistics and transport data

3. Route Optimization and Vehicle Routing Problems

- Shortest path algorithms (Dijkstra, A*)
- TSP, CVRP, VRPTW, and their variants
- Heuristics vs. exact solvers vs. metaheuristics (e.g., genetic algorithms)

4. Network Modeling and Flow Analysis

- Transport network representations: nodes, arcs, flows
- Multicommodity flows and network resilience
- Centrality, bottlenecks, and accessibility in logistics networks
- Tools: NetworkX, Gephi, Graphviz

5. Freight Demand Modeling and Forecasting

- Disaggregate vs. aggregate demand models
- Regression, clustering, and time series for freight flows
- Origin-destination matrix estimation
- Simulation of demand under different scenarios

6. Geospatial Analytics for Logistics

- Working with geospatial data: GPS, shapefiles, OpenStreetMap
- Route mapping and visualization using Geopandas and Folium
- Zonal analysis, catchment areas, and warehouse siting
- Use cases: cold chain logistics, last-mile delivery

7. Last-Mile and Urban Transportation Analytics

- Last-mile delivery challenges and microhub solutions
- Congestion, emissions, and urban delivery trade-offs
- Demand-responsive transport and shared mobility logistics
- Case studies: Amazon, DHL, UPS, city logistics platforms

Literatur

- Crainic, T.G. & Laporte, G. (1998). Fleet Management and Logistics
- Daganzo, C. F. (2005). Logistics Systems Analysis
- Ghiani, G., Laporte, G., & Musmanno, R. (2013). Introduction to Logistics Systems Management
- Taniguchi, E. & Thompson, R. (2015). City Logistics: Mapping the Future



Business Analytics

Fuerth

Modulbeschreibung

Retail and E-commerce Analytics

Modulnummer	Niveau Master	Studiengangsleitung Prof. Dr. Clemens Werkmeister
ECTS (Benotung) 6 (benotet)	Dauer Semester	Häufigkeit 2 per Year
Sprache English	Workload Insgesamt: 150 Stunden - davon Kontaktzeit: 34 Stunden - davon Selbststudium: 116 Stunden - davon Praxis: 0 Stunden	Veranstaltungstyp Interaktives Seminar Präsentation Übung

Teilnahmevoraussetzungen

Kursinformation

Modulprüfung

Prüfungsteile	Gewichtung (Bewertung)
Präsentation	100%

Qualifikationsziele (Learning Outcomes) und Kompetenzen

Nach erfolgreichem Abschluss des Moduls können die Studierenden ...

Handlungskompetenz

Students will be able to understand, analyze, and optimize retail and e-commerce processes using data science tools. They will gain the competence to develop data-driven strategies to enhance customer experience, logistics, and profitability in physical and digital retail environments.

Fachkompetenz „Wissen und Verstehen“

Students understand the structure, operations, and trends in modern retail and e-commerce ecosystems. They are able to evaluate and apply key metrics and technologies relevant to retail performance, customer interaction, and digital transformation.

Methodenkompetenz „Einsatz, Anwendung und Erzeugung von Wissen“

Students learn to apply analytical techniques such as market basket analysis, customer segmentation, and pricing optimization. They gain hands-on experience with tools and platforms used in retail analytics and e-commerce data tracking.

Selbstkompetenz „Wissenschaftliches Selbstverständnis und Professionalität“

Students strengthen their ability to manage complex analytical tasks independently, integrating business understanding with technical skills. They reflect critically on digital retail models and their implications for customer privacy, data usage, and business ethics.

Sozialkompetenz „Kommunikation und Kooperation“

Students collaborate on group projects simulating real-world retail and e-commerce scenarios. They improve their communication skills by presenting data-driven recommendations to varied stakeholders in a business context.

Constructive Alignment

The module aligns learning outcomes with teaching methods that combine theoretical foundations with practical analysis tasks. Assessments include team-based projects and individual assignments, ensuring students can apply knowledge to analyze real retail and e-commerce data, derive insights, and present strategic solutions.

Inhalt des Moduls

- Overview of retail and e-commerce ecosystems
- Omnichannel strategies and customer journeys
- Key retail metrics (e.g., conversion rates, average basket size, churn rate)
- Tools for e-commerce and retail data collection (e.g., Google Analytics, POS systems)
- Personalization, recommendation systems, and dynamic pricing
- Inventory and supply chain optimization in retail
- Customer segmentation and targeting
- Fraud detection and cybersecurity in e-commerce
- Mobile commerce and social commerce trends
- Ethical and legal considerations in digital commerce

Literatur

- Chaffey, D. (2022). Digital Marketing: Strategy, Implementation and Practice (8th ed.). Pearson.
- Grewal, D., Roggeveen, A. L., & Nordfält, J. (2017). The Future of Retailing. Emerald Group Publishing.
- Laudon, K. C., & Traver, C. G. (2021). E-Commerce 2021: Business, Technology and Society (16th ed.). Pearson.
- McKinsey & Company (2025). The Future of Retail: Annual Update. (Whitepaper)
- Kumar, V., & Reinartz, W. (2016). Creating Enduring Customer Value. Journal of Marketing, 80(6), 36–68.

Modulbeschreibung

Performance and Engagement Analytics

Modulnummer	Niveau Master	Studiengangsleitung Prof. Dr. Clemens Werkmeister
ECTS (Benotung) 6 (benotet)	Dauer Semester	Häufigkeit 2 per Year
Sprache English	Workload Insgesamt: 150 Stunden - davon Kontaktzeit: 34 Stunden - davon Selbststudium: 116 Stunden - davon Praxis: 0 Stunden	Veranstaltungstyp Fallarbeit Fallbearbeitung und Reflexion Gruppenarbeit Lehrvortrag Übung

Teilnahmevoraussetzungen

Kursinformation

Modulprüfung

Prüfungsteile	Gewichtung (Bewertung)
Studienarbeit	100%

Qualifikationsziele (Learning Outcomes) und Kompetenzen

Nach erfolgreichem Abschluss des Moduls können die Studierenden ...

Handlungskompetenz

Students will be able to analyze human capital data to support evidence-based decision-making in the areas of talent acquisition, employee engagement, and performance management. They will understand how to apply data-driven insights to improve organizational effectiveness and employee outcomes.

Fachkompetenz „Wissen und Verstehen“

Students understand the key concepts and frameworks related to talent management, workforce engagement, and performance measurement in organizations. They can evaluate and apply HR analytics to support strategic and operational decisions in human resource development.

Methodenkompetenz „Einsatz, Anwendung und Erzeugung von Wissen“

Students gain the ability to use tools such as regression analysis, machine learning, and sentiment analysis to assess employee performance, engagement levels, and turnover risk. They learn to work with HR datasets and dashboards to derive actionable insights.

Selbstkompetenz „Wissenschaftliches Selbstverständnis und Professionalität“

This module aligns theoretical understanding of talent and performance management with practical data analytics exercises. Learning is assessed through individual assignments and group projects involving real or simulated HR datasets, ensuring students can apply analytic techniques to support people-centric business decisions.

Sozialkompetenz „Kommunikation und Kooperation“

Students enhance their ability to collaborate across departments and communicate sensitive insights effectively and ethically. They practice empathy and active listening in the context of interpreting and discussing employee-related data and trends.

Constructive Alignment

This module aligns theoretical understanding of talent and performance management with practical data analytics exercises. Learning is assessed through individual assignments and group projects involving real or simulated HR datasets, ensuring students can apply analytic techniques to support people-centric business decisions.

Inhalt des Moduls

- Fundamentals of talent management and organizational behavior
- Employee lifecycle: recruitment, retention, development
- Engagement drivers and measurement methods (e.g., surveys, pulse checks)
- Performance management systems and KPIs
- HR analytics: metrics, models, and tools
- Predictive modeling of turnover and performance outcomes
- Natural Language Processing (NLP) in engagement analysis
- Diversity, equity, and inclusion analytics
- Legal and ethical considerations in people analytics
- Communicating HR insights to stakeholders

Literatur

- Cascio, W. F., & Boudreau, J. W. (2010). Investing in People: Financial Impact of Human Resource Initiatives (2nd ed.). Pearson.
- Fitz-Enz, J., & Mattox, J. R. (2014). Predictive Analytics for Human Resources. Wiley.
- Ulrich, D., Brockbank, W., Johnson, D., Sandholtz, K., & Younger, J. (2012). HR Competencies: Mastery at the Intersection of People and Business. Society for Human Resource Management.
- Bersin, J. (2023). Irresistible: The Seven Secrets of the World's Most Enduring, Employee-Focused Organizations. Ideapress Publishing.
- Weber, B. (2013): People Analytics. FT Press.
- Ferrar, J.; Green, D. (2021): Excellence in People Analytics. Kogan Page.
- McNulty, K. (2021): Handbook of Regression Modeling in People Analytics. With Examples in R and Python. Chapman and Hal/CRC Press.



AI-Driven Bioinformatics & Life Sciences

Munich

Modulbeschreibung

AI for Drug Discovery

Modulnummer	Niveau Master	Studiengangsleitung Prof. Dr. Swati Chandna
ECTS (Benotung) 6 (benotet)	Dauer Semester	Häufigkeit 2 per Year
Sprache English	Workload Insgesamt: 150 Stunden - davon Kontaktzeit: 34 Stunden - davon Selbststudium: 116 Stunden - davon Praxis: 0 Stunden	Veranstaltungstyp Gruppenarbeit Lehrvortrag Projektarbeit Übung

Teilnahmevoraussetzungen

Kursinformation

Modulprüfung

Prüfungsteile	Gewichtung (Bewertung)
Projektarbeit	100%

Qualifikationsziele (Learning Outcomes) und Kompetenzen

Nach erfolgreichem Abschluss des Moduls können die Studierenden ...

Handlungskompetenz

This module introduces students to the application of artificial intelligence and machine learning techniques in the field of drug discovery and development. Students will explore data types such as molecular graphs, omics datasets, and high-throughput screening outputs. Emphasis is placed on applying supervised learning, generative models, and deep learning architectures to tasks such as virtual screening, de novo drug design, and target prediction. The module also addresses challenges in model interpretability, reproducibility, and translational relevance in biomedical research.

Fachkompetenz „Wissen und Verstehen“

- Understand the stages of the drug discovery pipeline and identify AI-amenable tasks
- Learn the fundamentals of chemical representations (SMILES, molecular graphs, descriptors)
- Apply ML and DL models to predict bioactivity, ADMET properties, and molecular similarity
- Interpret QSAR (Quantitative Structure–Activity Relationship) modeling results in pharmacological contexts

Methodenkompetenz „Einsatz, Anwendung und Erzeugung von Wissen“

- Preprocess and featurize molecular and biological datasets (e.g., ChEMBL, PubChem, UniProt)
- Apply regression, classification, and clustering algorithms for activity prediction
- Use graph neural networks (GNNs), transformers, and generative models (e.g., VAEs) for molecular tasks
- Evaluate models using appropriate bioinformatics metrics (e.g., ROC-AUC, precision@k, enrichment factor)

Selbstkompetenz „Wissenschaftliches Selbstverständnis und Professionalität“

- Reflect on the ethical, legal, and societal implications of AI in drug discovery
- Assess the reproducibility and generalizability of AI results in health contexts
- Demonstrate initiative in problem-solving and critical model evaluation

Sozialkompetenz „Kommunikation und Kooperation“

- Collaborate across disciplines (biology, chemistry, data science) on joint projects
- Communicate modeling results and limitations clearly to domain experts
- Address stakeholder concerns in biomedical R&D settings (e.g., model trust, risk, safety)

Constructive Alignment

- Lectures and demos provide foundational and advanced understanding of molecular modeling and AI (Specialist)
- Hands-on labs and coding exercises enable application of ML/DL tools to real bioactivity datasets (Methodological)
- Team projects simulate interdisciplinary collaboration in drug discovery R&D (Social)
- Ethics workshops and reflections build awareness of the limitations and responsibilities of biomedical AI (Personal)

Inhalt des Moduls

- Introduction to Drug Discovery and the Role of AI
- Bioactivity Prediction and QSAR Modeling
- Deep Learning for Molecules
- Generative Models in Drug Discovery
- Target Prediction and Polypharmacology
- Data Sources and Tools for AI in Drug Discovery
- Model Evaluation, Validation, and Deployment
- Ethics, Safety, and Explainability in Biomedical AI

Literatur

- Chen, H., Engkvist, O., Wang, Y., Olivecrona, M., & Blaschke, T. (2018). The Rise of Deep Learning in Drug Discovery – Drug Discovery Today
- Vamathevan, J. et al. (2019). Applications of Machine Learning in Drug Discovery and Development – Nature Reviews Drug Discovery
- Altae-Tran, H. et al. (2017). Low Data Drug Discovery with One-Shot Learning – ACS Central Science
- Bender, A., & Cortes-Ciriano, I. (2021). Artificial Intelligence in Drug Discovery: What Is Realistic, What Are Illusions? – Part of Drug Discovery Today

Modulbeschreibung

Biomedical image analysis

Modulnummer	Niveau Master	Studiengangsleitung Prof. Dr. Swati Chandna
ECTS (Benotung) 6 (benotet)	Dauer Semester	Häufigkeit 2 per Year
Sprache English	Workload Insgesamt: 150 Stunden - davon Kontaktzeit: 34 Stunden - davon Selbststudium: 116 Stunden - davon Praxis: 0 Stunden	Veranstaltungstyp Diskussion Gruppenarbeit

Teilnahmevoraussetzungen

Kursinformation

Modulprüfung

Prüfungsteile	Gewichtung (Bewertung)
Praktische Arbeit	100%

Qualifikationsziele (Learning Outcomes) und Kompetenzen

Nach erfolgreichem Abschluss des Moduls können die Studierenden ...

Handlungskompetenz

This module introduces students to the field of biomedical image analysis, focusing on how computer vision, machine learning, and deep learning techniques are applied to extract meaningful information from medical and biological imaging data. Students will work with modalities such as MRI, CT, histopathology, and microscopy, using algorithms for preprocessing, segmentation, classification, and anomaly detection. The course emphasizes both technical skills and ethical considerations in clinical and biomedical research contexts.

Fachkompetenz „Wissen und Verstehen“

- Understand medical image modalities (MRI, CT, PET, histology) and their diagnostic relevance
- Apply image preprocessing, segmentation, and enhancement techniques to biomedical data
- Evaluate classification, detection, and annotation pipelines in a biomedical context

Assess image quality, variability, and annotation reliability in medical datasets

Methodenkompetenz „Einsatz, Anwendung und Erzeugung von Wissen“

- Use tools like OpenCV, scikit-image, SimpleITK, and PyTorch/TensorFlow for image analysis
- Implement pipelines for denoising, filtering, morphological operations, and registration
- Train and evaluate deep learning models (e.g., CNNs, U-Net, ResNet) on biomedical images
- Apply evaluation metrics (Dice coefficient, IoU, sensitivity/specificity)

Selbstkompetenz „Wissenschaftliches Selbstverständnis und Professionalität“

- Reflect on the ethical and legal implications of medical imaging AI (e.g., bias, privacy, fairness)
- Demonstrate accountability in handling sensitive patient and clinical data
- Critically evaluate limitations of algorithms in high-stakes biomedical decision-making

Sozialkompetenz „Kommunikation und Kooperation“

- Collaborate in interdisciplinary teams with biologists, radiologists, and data scientists
- Communicate model results to clinicians and biomedical stakeholders
- Address issues of clinical trust, explainability, and medical safety

Constructive Alignment

- Lectures and tool demos introduce theory, modalities, and workflows (Specialist)
- Labs and coding assignments build applied skills in biomedical image preprocessing and modeling (Methodological)
- Group projects and presentations encourage interdisciplinary collaboration and clinical communication (Social)
- Ethics discussions and reflections foster awareness of fairness, safety, and data privacy (Personal)

Inhalt des Moduls

1. Introduction to Biomedical Imaging
 - Overview of imaging modalities (MRI, CT, PET, Ultrasound, X-ray, Histology)
 - Biomedical data sources: NIH, TCIA, Allen Brain Atlas, CAMELYON
 - DICOM format and metadata interpretation
2. Image Preprocessing and Enhancement
 - Image filtering, normalization, histogram equalization
 - Noise reduction (Gaussian, median filtering)
 - Contrast enhancement and standardization
 - Image registration and alignment techniques
3. Image Segmentation Techniques
 - Thresholding and region growing
 - Watershed segmentation
 - Machine learning-based segmentation
 - Deep learning with U-Net and variants
4. Feature Extraction and Classification
 - Texture, shape, intensity-based features
 - Feature selection and dimensionality reduction (PCA, t-SNE)
 - Classification using traditional ML (SVM, Random Forest, k-NN)
 - CNN-based classification in histopathology and radiology
5. Deep Learning for Biomedical Imaging
 - CNNs for image classification and detection
 - U-Net, V-Net, and 3D CNNs for medical image segmentation
 - Transfer learning and data augmentation
 - Performance evaluation and medical model calibration
6. Multi-Modal and 3D Imaging
 - 3D imaging data handling and visualization
 - Multimodal fusion (e.g., MRI + genomics, PET + CT)
 - Volumetric analysis and reconstruction
7. Annotation, Datasets, and Weak Supervision
 - Ground truth generation and inter-observer variability
 - Semi-supervised and weakly supervised learning
 - Dealing with label noise and small datasets
8. Explainability and Model Validation
 - Saliency maps, Grad-CAM, SHAP for imaging models
 - Cross-validation, bootstrapping, and external validation
 - Model uncertainty and clinical decision support

Literatur

- Gonzalez & Woods. Digital Image Processing (Medical extensions)
- Zhou, S. K., Greenspan, H., & Shen, D. (2019). Deep Learning for Medical Image Analysis

-Shilo, S., & Benjamini, Y. (2022). Statistical Challenges in Modern Medical Imaging



Semester 3: Elective Semester

3. Semester/Internship/Semester Abroad

Modulbeschreibung

Internship/Semester Abroad

Modulnummer	Niveau Master	Studiengangsleitung Prof. Dr. Swati Chandna
ECTS (Benotung) 30 (benotet)	Dauer Semester	Häufigkeit
Sprache English	Workload Insgesamt: 750 Stunden - davon Kontaktzeit: 0 Stunden - davon Selbststudium: 0 Stunden - davon Praxis: 0 Stunden	Veranstaltungstyp N/A

Teilnahmevoraussetzungen

Completion of foundational modules in Python, statistics, machine learning, and data engineering

Kursinformation

Modulprüfung

Prüfungsteile	Gewichtung (Bewertung)
Präsentation	30%
Praxisbericht	50%
Lerntagebuch	20%

Qualifikationsziele (Learning Outcomes) und Kompetenzen

Nach erfolgreichem Abschluss des Moduls können die Studierenden ...

Handlungskompetenz

This applied module gives students the opportunity to extend their learning beyond the classroom through either a professional internship or an academic semester abroad. Both paths are designed to support the application of data science and AI knowledge in authentic, interdisciplinary, and cross-cultural environments, fostering personal, professional, and methodological growth.

Fachkompetenz „Wissen und Verstehen“

- Apply advanced data science techniques (e.g., machine learning, data engineering, AI modeling) in authentic problem-solving contexts
- Deepen subject-matter expertise in Semester 3 themes such as Deep Learning, Cloud Computing, Data Visualization, Entrepreneurship, or domain-specific AI (e.g., health, media, logistics)
- Translate domain-specific needs into data-driven, deployable solutions or academically rigorous projects

Methodenkompetenz „Einsatz, Anwendung und Erzeugung von Wissen“

- Operate independently within a data-intensive workflow (ETL, modeling, evaluation, visualization)
- Learn new tools, workflows, or academic methods as used in industry or partner institutions
- Engage in project-based learning through real-world assignments, research, or case studies
- Reflect critically on the data lifecycle and implications of applied models

Selbstkompetenz „Wissenschaftliches Selbstverständnis und Professionalität“

- Demonstrate initiative, self-direction, and resilience in ambiguous or new environments
- Reflect on personal learning progress and ethical responsibility in applied AI contexts
- Strengthen adaptability through industry practices or academic mobility
- Gain clarity on career or research direction through lived experience

Sozialkompetenz „Kommunikation und Kooperation“

- Collaborate in interdisciplinary and multicultural teams
- Communicate clearly with non-technical stakeholders or academic peers
- Present work results, explain methodologies, and defend technical decisions effectively
- Develop global and professional networks for career development

Constructive Alignment

- The module links theory and application through authentic immersion, whether in the workplace or a global academic environment.
- Students engage with real projects, stakeholders, or research, translating classroom knowledge into practical value.
- Learning outcomes are achieved through contextualized action, continuous feedback, and final reflection, as aligned with the CORE model.
- The dual-path flexibility ensures that all students, regardless of career goals, can demonstrate independent problem-solving, social competence, and self-efficacy in applied data science.
- Assessment reflects growth, contribution, and competence acquisition—not only content mastery.

Inhalt des Moduls

1. Path A: 6 Months Internship in Data Science and AI

- Students work in a professional setting (company, research lab, tech startup, NGO) where they contribute to data-driven projects. Tasks may include:
- Data preparation and ETL workflows
- Predictive modeling or forecasting
- Deployment of AI tools or APIs
- Dashboard development or reporting pipelines
- ML/AI engineering or MLOps
- Exposure to agile, DevOps, or business intelligence practices

Deliverables:

- Internship confirmation letter
- Internship report (technical + reflective)
- Presentation or oral defense
- Employer evaluation

2. Path B: Semester Abroad

- Students study abroad at a partner university, selecting courses that align with at least three of the following focus areas:
- Deep Learning and AI Engineering
- Cloud Computing and Distributed Systems
- Data Visualization and Storytelling
- Innovation and Entrepreneurship
- Applied AI in Health, Media, Environment, or Logistics

Academic activities may include coursework, project seminars, research internships, or hackathons. Courses must be approved via a Learning Agreement.

Deliverables:

- Learning Agreement (pre-departure)
- Transcript of Records (post-semester)
- Reflection report (3–5 pages)
- Final presentation or oral interview

Literatur

- Joel Grus (2019). Data Science from Scratch (2nd ed.)
- Emily Robinson & Jacqueline Nolis (2020). Build a Career in Data Science
- Harvard Business Review Editors (2018). HBR Guide to Data Analytics Basics for Managers

Modulbeschreibung

Deep Learning

Modulnummer	Niveau Master	Studiengangsleitung Prof. Dr. Swati Chandna
ECTS (Benotung) 6 (benotet)	Dauer Semester	Häufigkeit 2 per Year
Sprache English	Workload Insgesamt: 150 Stunden - davon Kontaktzeit: 64 Stunden - davon Selbststudium: 86 Stunden - davon Praxis: 0 Stunden	Veranstaltungstyp Lehrvortrag Tutorium Übung

Teilnahmevoraussetzungen

Kursinformation

Modulprüfung

Prüfungsteile	Gewichtung (Bewertung)
Praktische Arbeit	100%

Qualifikationsziele (Learning Outcomes) und Kompetenzen

Nach erfolgreichem Abschluss des Moduls können die Studierenden ...

Handlungskompetenz

This module aims to provide students with a comprehensive understanding of deep learning principles, architectures, and applications. Students will develop the theoretical knowledge and practical skills necessary to design, train, and evaluate deep learning models for a variety of tasks, including image recognition, natural language processing, and generative modeling.

Fachkompetenz „Wissen und Verstehen“

Students who complete this module will be able to:

- Explain the fundamental concepts of deep learning, including its relationship to artificial intelligence and neural networks.
- Describe the architecture and operation of linear neural networks, including concepts like linear regression, loss functions, and gradient descent.
- Understand multilayer neural networks, including feed-forward networks, activation functions, and techniques for preventing network collapse.
- Explain the backpropagation algorithm and its role in training deep neural networks, including challenges like vanishing/exploding gradients.
- Describe techniques for improving deep neural networks, such as regularization, dropout, batch normalization, and data augmentation.
- Understand the architecture and applications of convolutional neural networks (CNNs) for image processing tasks.
- Explain the principles and applications of recurrent neural networks (RNNs) for processing sequential data, including challenges and modern variants like LSTMs and GRUs.
- Describe advanced deep learning concepts, including transfer learning, autoencoders, and graph neural networks (GANs).

Methodenkompetenz „Einsatz, Anwendung und Erzeugung von Wissen“

Through practical exercises and projects, students will gain the ability to:

- Implement and train linear and multilayer neural networks using deep learning frameworks (e.g., TensorFlow, PyTorch).
- Apply backpropagation and optimization algorithms to train deep learning models effectively.
- Design and implement convolutional neural networks for image classification and object detection tasks.
- Develop and train recurrent neural networks for sequence-to-sequence learning and other sequential data processing tasks.
- Utilize transfer learning techniques to adapt pre-trained models to new tasks and datasets.
- Implement and train autoencoders for dimensionality reduction and feature extraction.
- Develop and train generative adversarial networks for generating new data samples.
- Evaluate the performance of deep learning models using appropriate metrics and techniques.

Selbstkompetenz „Wissenschaftliches Selbstverständnis und Professionalität“

This course is designed to help students:

- Independently research and stay up-to-date with the latest advancements in deep learning.
- Critically evaluate the strengths and limitations of different deep learning architectures and techniques.
- Manage time and resources effectively when working on deep learning projects.
- Develop a critical and reflective approach to problem-solving in the context of deep learning.
- Demonstrate ethical considerations in the application of deep learning technologies.

Sozialkompetenz „Kommunikation und Kooperation“

This module will encourage the development of the following social skills:

- Communicate complex technical concepts and results clearly and concisely to both technical and non-technical audiences.
- Present project findings and defend design choices in a professional and persuasive manner.
- Engage in constructive discussions during class and Q&A.
- Seek and incorporate feedback from instructors.

Constructive Alignment

The learning objectives, course content, and assessment methods in this module are carefully aligned to ensure a coherent and effective learning experience. The module combines theoretical lectures with hands-on practical sessions. The primary assessment is Practical Work, requiring students to demonstrate mastery of deep learning concepts and techniques by improving upon existing research:

- Research Paper Selection: Identify a peer-reviewed neural network research paper that focuses on a specific deep learning task and utilizes a publicly available dataset.
- Dataset Understanding and Preparation: Obtain, analyze, and prepare the chosen dataset, documenting their understanding and any preprocessing steps.
- Baseline Model Replication: Implement the method/model from the selected paper to establish a baseline for comparison, aiming to replicate the published results.
- Model/Method Improvement: Develop and implement an improved model/method, striving to demonstrably outperform the baseline through various techniques.
- Evaluation: Evaluate both baseline and improved models/methods using appropriate metrics and statistical significance testing on a held-out test set.
- Documentation: Prepare a detailed report documenting the entire project, including paper summary, dataset analysis, model descriptions, results, and critical discussion.
- Code Submission: Submit well-documented, clean, and reproducible code for both the baseline and improved implementations.

Inhalt des Moduls

1. Foundations of Deep Learning

- Motivation and history of neural networks
- Perceptron, universal approximation theorem
- Representational power of deep networks
- Introduction to key challenges: vanishing gradients, overfitting, generalization

2. Linear Neural Networks

- Linear regression as a neural network
- Mean squared error and gradient descent
- Multi-input and multi-output regression
- Weight initialization and batch processing

3. Multilayer Neural Networks

- Architecture of feedforward networks
- Activation functions: sigmoid, tanh, ReLU, GELU
- Forward propagation and computation graphs
- Vectorization and computational efficiency

4. Backward Propagation

- Derivation of backpropagation
- Chain rule and gradient computation
- Computational graph frameworks (TensorFlow, PyTorch autograd)
- Hands-on implementation of backprop for a multilayer perceptron (MLP)

5. Improving Deep Neural Networks

- Loss functions: cross-entropy, hinge loss, custom losses
- Regularization: dropout, weight decay (L2), early stopping
- Batch normalization and layer normalization
- Hyperparameter tuning: learning rate schedules, optimizers (SGD, Adam)

6. Convolutional Neural Networks (CNNs)

- Convolutional layers, kernels, padding, stride
- Pooling layers, parameter sharing
- CNN architectures: LeNet, AlexNet, VGG, ResNet
- Applications in image classification, object detection

7. Transfer Learning

- Fine-tuning vs. feature extraction
- Pre-trained models: ImageNet, HuggingFace Transformers
- Domain adaptation and model reusability
- Case studies in medical imaging, document analysis, small data learning

8. Recurrent Neural Networks (RNNs)

- Sequential modeling and temporal dependencies
- Basic RNN cell and time-unfolded networks
- Backpropagation through time (BPTT)

- Use cases: sentiment analysis, sequence labeling, time-series prediction

9. Modern RNNs: LSTM and GRU

- Long Short-Term Memory: gates and architecture
- GRU vs. LSTM trade-offs
- Exploding/vanishing gradients revisited
- Use in language modeling and video processing

10. Autoencoders and Representation Learning

- Autoencoder architecture: encoder, bottleneck, decoder
- Denoising autoencoders and regularized AEs
- Variational Autoencoders (VAEs): generative modeling and latent space
- Applications in anomaly detection, compression, data denoising

11. Graph Neural Networks (GNNs) ☒

- Introduction to graph-structured data
- Node, edge, and graph-level tasks
- Message Passing Neural Networks (MPNN), GCN, GAT

Literatur

- Goodfellow, I., Bengio, Y., & Courville, A. (2016). Deep Learning. MIT Press.
- Nielsen, M. A. (2015). Neural Networks and Deep Learning. Determination Press.
- Geron, A. (2019). Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow (2nd ed.). O'Reilly Media.
- Zhang, A., Lipton, Z. C., Li, M., & Smola, A. J. (2021). Dive into Deep Learning. Available online.
- Bishop, C. M. (2006). Pattern Recognition and Machine Learning. Springer.
- Sutton, R. S., & Barto, A. G. (2018). Reinforcement Learning: An Introduction (2nd ed.). MIT Press.
- Chollet, F. (2017). Deep Learning with Python. Manning Publications.
- Vajjala, S., Majumder, B., & Poria, S. (2020). Deep Learning for Natural Language Processing. Springer
- Li, Fei-Fei, and Andrej Karpathy, and Justin Johnson. CS231n: Convolutional neural networks for visual recognition. (2016).

Modulbeschreibung

Cloud Computing and Data Architectures

Modulnummer	Niveau Master	Studiengangsleitung Prof. Dr. Swati Chandna
ECTS (Benotung) 6 (benotet)	Dauer Semester	Häufigkeit 2 per Year
Sprache English	Workload Insgesamt: 150 Stunden - davon Kontaktzeit: 45 Stunden - davon Selbststudium: 105 Stunden - davon Praxis: 0 Stunden	Veranstaltungstyp Gruppenarbeit Tutorium Vorlesung

Teilnahmevoraussetzungen

Kursinformation

Modulprüfung

Prüfungsteile	Gewichtung (Bewertung)
Projektarbeit	100%

Qualifikationsziele (Learning Outcomes) und Kompetenzen

Nach erfolgreichem Abschluss des Moduls können die Studierenden ...

Handlungskompetenz

This module aims to equip students with the knowledge and practical skills to design, implement, and manage data-intensive applications and systems on cloud platforms. Students will learn about various data architectures, cloud deployment models, distributed data processing frameworks, and specific cloud services, with a focus on hands-on implementation and collaborative project work.

Fachkompetenz „Wissen und Verstehen“

Graduates of this module will be able to:

- Compare and contrast different data architectures, including Data Warehouses, Data Lakes, Lambda architecture, and Kappa architecture.
- Understand the principles and practices of containerization using Docker and container orchestration using Kubernetes.
- Explain the concepts of distributed data processing and utilize frameworks like Apache Spark and PySpark.
- Describe the functionality and use cases of Databricks as a unified data analytics platform.
- Understand the principles of real-time data streaming and utilize technologies like Apache Kafka.
- Describe the core services of Google Cloud Platform (GCP), including BigQuery, Cloud Storage, App Engine, Dataflow, Pub/Sub, and Dataproc.
- Differentiate various options of data storage in the cloud.
- Understand the difference between Batch and Stream processing.

Methodenkompetenz „Einsatz, Anwendung und Erzeugung von Wissen“

Upon completion of this module, students will be able to:

- Design and implement data pipelines using appropriate cloud services and data architectures.
- Deploy and manage applications using Docker and Kubernetes.
- Develop and execute data processing jobs using Apache Spark and PySpark.
- Utilize Databricks for data exploration, analysis, and machine learning.
- Implement real-time data streaming applications using Apache Kafka.
- Utilize GCP services (BigQuery, Cloud Storage, etc.) for data storage, processing, and analysis.
- Design and implement serverless applications using Google Cloud App Engine and Dataflow.
- Configure and manage distributed data processing clusters using Google Cloud Dataproc.

Selbstkompetenz „Wissenschaftliches Selbstverständnis und Professionalität“

This module encourages students to:

- Independently research and evaluate different cloud computing technologies and data architectures.
- Critically assess the trade-offs between different cloud service models and deployment options.
- Manage time and resources effectively when working on cloud-based projects.
- Adapt to the rapidly evolving landscape of cloud computing technologies.
- Take ownership of project deliverables and demonstrate accountability for their work within a team.

Sozialkompetenz „Kommunikation und Kooperation“

This module will emphasize the development of teamwork and communication:

- Effectively collaborate with other members within the group.
- Present solutions and justify technical decisions.
- Communicate technical concepts, and provide constructive feedback.
- Coordinate tasks, manage timelines, and resolve conflicts within the group.

Constructive Alignment

The learning objectives, course content, and assessment are tightly integrated to provide a comprehensive learning experience. The module combines theoretical lectures with extensive hands-on labs and a significant group project. The assessment, Project Work (in groups of 3), will require students to design, implement, and deploy a complete data-intensive application on a cloud platform (likely GCP, given the course content). The project will be evaluated based on:

- Architecture Design: Students must select and justify an appropriate data architecture (Data Warehouse, Data Lake, Lambda, or Kappa) based on the project requirements. The design should clearly articulate the rationale for the chosen architecture and its components.
- Cloud Service Selection: Students must choose appropriate cloud services (from GCP and potentially other providers) for data storage, processing, streaming, and application deployment. The selection should be justified based on factors like cost, scalability, performance, and ease of use.
- Implementation: Students must implement the designed architecture and application using the chosen technologies (Docker, Kubernetes, Spark, Kafka, Databricks, GCP services). The implementation should be well-documented, robust, and scalable.
- Data Processing and Analysis: The application should demonstrate effective data processing and analysis capabilities, relevant to the chosen project scenario.
- Deployment: The application should be successfully deployed and managed on the cloud platform, demonstrating proficiency with containerization and orchestration.
- Teamwork and Collaboration: The project will be assessed on the effectiveness of the group's collaboration, including task division, communication, conflict resolution, and overall teamwork. Evidence of equal contribution from all group members is expected.
- Documentation and Presentation: The group must prepare a comprehensive report documenting the project's design, implementation, and results. They will also present their project to the class, demonstrating the functionality of the application and answering questions.

Inhalt des Moduls

1. Introduction to Cloud Data Architectures

- Role of cloud in modern data engineering and analytics
- Overview of cloud service models: IaaS, PaaS, SaaS
- Data architectures in context: OLAP vs OLTP, batch vs stream

2. Modern Data Architecture Patterns

- Data Warehouse vs Data Lake
- Architecture, use cases, and limitations
- Lambda Architecture
- Kappa Architecture
- Real-time-first design and stream reprocessing

3. Distributed Computing with Apache Spark and PySpark

- Introduction to Spark architecture: driver, executors, DAGs
- Core concepts: RDD, DataFrame, lazy evaluation
- Data wrangling and transformation with PySpark
- Optimizations: partitioning, caching, Spark SQL

4. Real-Time Data Pipelines with Apache Kafka

- Event streaming and pub/sub concepts
- Kafka architecture: brokers, partitions, offsets, topics
- Producers, consumers, and message retention

5. Containerization and Cloud Deployment

- Docker: containers, images, Dockerfiles, and registries
- Kubernetes fundamentals: pods, deployments, services
- Scaling analytics pipelines with Kubernetes
- Orchestration in real-time and batch data pipelines

6. Introduction to Databricks

- Unified analytics for batch and stream
- Collaborative notebooks, Delta Lake, and ML integration
- Use cases: large-scale ETL, ML model training, experimentation

7. Google Cloud Platform (GCP) Overview

- GCP project structure and billing
- Authentication, IAM roles, and service accounts
- Introduction to key services for data engineers

8. GCP Data Storage & Querying Tools

- Google Cloud Storage (GCS): object storage, buckets, lifecycle policies
- Google BigQuery: data warehouse-as-a-service, SQL for big data
- BigQuery use cases: logs, transactions, public datasets
- Cost modeling and optimization strategies

9. GCP Data Orchestration and Compute

- Google App Engine: scalable applications without managing infrastructure

- Google Dataflow: unified batch/stream processing with Apache Beam
- Pipelines for ETL, log processing, sensor streams
- Comparison with other orchestration platforms (e.g., Airflow)

10. GCP Streaming and Processing Engines

- Google Pub/Sub: scalable message ingestion for event-driven systems
- Architecture, delivery guarantees, dead-letter topics
- Google Dataproc: managed Hadoop/Spark cluster
- Running distributed Spark jobs with GCP-native integrations

Literatur

- Marz, N., & Warren, J. (2015). Big Data: Principles and best practices of scalable realtime data systems. Manning Publications.
- Kreps, J. (2014). I Heart Logs: Event Data, Stream Processing, and Data Integration. O'Reilly Media.
- Chambers, B., & Zaharia, M. (2018). Spark: The Definitive Guide: Big Data Processing Made Simple. O'Reilly Media.
- Narkhede, N., Shapira, G., & Palino, T. (2017). Kafka: The Definitive Guide. O'Reilly Media.
- Burns, B., Beda, J., & Hightower, K. (2017). Kubernetes: Up and Running: Dive into the Future of Infrastructure. O'Reilly Media.
- Luksa, M. (2017). Kubernetes in Action. Manning Publications.
- Raj, G. B. (2017). Google Cloud Platform Cookbook. Packt Publishing.
- Lesniak, D., Regunath, M., and Wirfs, J. (2022) Google Cloud Certified Professional Data Engineer Study Guide. Sybex.

Modulbeschreibung

Data Visualization and Storytelling

Modulnummer	Niveau Master	Studiengangsleitung Prof. Dr. Swati Chandna
ECTS (Benotung) 6 (benotet)	Dauer Semester	Häufigkeit 2 per Year
Sprache English	Workload Insgesamt: 150 Stunden - davon Kontaktzeit: 48 Stunden - davon Selbststudium: 102 Stunden - davon Praxis: 0 Stunden	Veranstaltungstyp Gruppenarbeit Lehrvortrag Seminar Übung

Teilnahmevoraussetzungen

Kursinformation

Modulprüfung

Prüfungsteile	Gewichtung (Bewertung)
Projektarbeit	100%

Qualifikationsziele (Learning Outcomes) und Kompetenzen

Nach erfolgreichem Abschluss des Moduls können die Studierenden ...

Handlungskompetenz

This module introduces students to the principles and practices of designing clear, compelling, and ethical data visualizations. It focuses on transforming complex datasets into engaging visual stories using tools such as Tableau, Power BI, and Plotly Dash. Students will learn how to align visuals with the needs of different audiences, develop interactive dashboards, and apply storytelling techniques that support data-driven decision-making in various professional contexts.

Fachkompetenz „Wissen und Verstehen“

- Understand the role of data visualization in analysis, insight communication, and persuasion
- Choose appropriate visualization types based on data structure and audience context
- Structure narrative flows for exploration (dashboards), explanation (reports), and persuasion (presentations)
- Evaluate the effectiveness and accuracy of visual encodings

Methodenkompetenz „Einsatz, Anwendung und Erzeugung von Wissen“

- Use Tableau and Power BI to build dynamic dashboards and business reports
- Create web-based interactive visualizations using Plotly Dash (Python)
- Apply design best practices: layout, typography, color theory, and interactivity
- Integrate multiple datasets into visual narratives with filters, KPIs, and drill-downs

Selbstkompetenz „Wissenschaftliches Selbstverständnis und Professionalität“

- Reflect on how data stories influence public opinion and decision-making
- Recognize the ethical risks of misrepresentation, omission, or cognitive bias
- Design accessible and inclusive visualizations that consider diverse user needs

Sozialkompetenz „Kommunikation und Kooperation“

- Communicate insights to technical and non-technical audiences
- Collaborate on storytelling projects with domain experts and stakeholders
- Practice empathy and user-centered design in visualization choices
- They are capable of recognizing the needs of a target audience so that they prepare information and communicate insights properly.

Constructive Alignment

- Lectures and tool demonstrations equip students with design theory and software skills (Specialist + Methodological)
- Tool labs (Tableau, Power BI, Dash) provide hands-on experience in visualization design and deployment (Methodological)
- Group storytelling projects build collaboration and stakeholder communication (Social)
- Ethical case studies and reflections promote responsible, user-centered design awareness (Personal)

Inhalt des Moduls

1. Foundations of Visualization and Storytelling

- History and evolution of data visualization
- Visual perception: Gestalt principles, preattentive attributes, and cognitive load
- Data storytelling frameworks: exploratory vs explanatory, narrative arcs
- Ethical visualization: truthfulness, misleading graphs, visual bias
- Audience-centered communication and framing

2. Data Transformation and Visualization Strategy

- Data shaping for visuals: wide vs long format
- Aggregation, filtering, joining, and reshaping data
- Choosing the right chart for the task: comparison, distribution, relationship, composition
- Visual grammar: axis, scale, hierarchy, layout, and whitespace
- Use of color, typography, and iconography for emphasis and accessibility

3. Advanced Chart Types and Visual Patterns

- Time-series: area charts, confidence bands, horizon charts
- Multivariate plots: scatter matrix, parallel coordinates
- Hierarchical data: treemaps, sunbursts, waterfall charts
- Geospatial visualizations: symbol maps, choropleths, density maps
- Visualizing uncertainty and outliers

4. Tableau for Interactive Data Dashboards

- Data connectors and relationships
- Calculated fields, table calculations, sets, and LOD expressions
- Interactive filters, parameters, and dashboard actions
- Story points for sequencing narrative flows
- Exporting, publishing, and embedding dashboards

5. Power BI for Business Data Visualization

- Importing data from databases, Excel, APIs
- DAX basics for calculations and measures
- Designing slicers, KPI cards, and matrix visualizations
- Power BI Service: sharing, versioning, embedding
- AI visuals: Q&A, decomposition trees, key influencers

6. Dash (Plotly) for Interactive Web Visualizations

- Overview of Dash architecture: layout, components, and callbacks
- Using Plotly for rich, interactive visualizations (scatter, bar, choropleth, etc.)
- Building interactive controls: dropdowns, sliders, checklists
- Linking widgets to graphs using callbacks
- Modular app design and component reuse

7. Visual Storytelling and Presentation Design

- Narrative structures in data storytelling (hero's journey, pyramid principle)
- Annotating for impact: tooltips, highlights, guided interaction
- Combining visuals with voice, video, and slideware

-Presenting to executives, policymakers, or general audiences

8. Accessibility and Ethical Design in Data Visualization

- Designing for colorblindness, screen readers, mobile responsiveness
- Avoiding cognitive overload and misleading visuals
- Responsible representation: cultural sensitivity, neutrality, and privacy

Literatur

- Alexander B: The New Digital Storytelling: Creating Narratives with New Media, ABC-Clio, 2011.
- Berinato S: Good Charts: The HBR Guide to Making Smarter, More Persuasive Data Visualizations, Harvard Business Review Press, 2016.
- DeBarros A: Practical SQL: A Beginner's Guide to Storytelling with Data, No Starch Press, 2018.
- Evergreen SDH: Effective Data Visualization: The Right Chart for the Right Data, Sage Pubn, 2016.
- Foreman JW: Data Smart: Using Data Science to Transform Information into Insight, 1st edition, Wiley, 2013.
- Marr B: Big Data: Using SMART Big Data, Analytics and Metrics to Make Better Decisions and Improve Performance, 1st edition, Wiley, 2015.
- Nussbaumer Knafl C: Storytelling mit Daten. Die Grundlagen der effektiven Kommunikation und Visualisierung mit Daten, 1. Auflage, Vahlen, 2017.
- Provost F & Fawcett T: Data Science for Business: What you need to know about data mining and data-analytic thinking, 1st edition, O'Reilly, 2013.
- Wong DM: The Wall Street Journal Guide to Information Graphics: The Dos and Don'ts of Presenting Data, Facts, and Figures, Reprint- Auflage, Ww Norton & Co, 2014.
- Few S: Information Dashboard Design, Analytics Press, 2015
- Iliinsky N, Steele J: Beautiful Visualization, O'Reilly Media, 2010
- Cairo A: The functional art: An introduction to information graphics and visualization, New Riders Publishing, 2012
- Berinato S: Good Charts: The HBR Guide to Making Smarter, More Persuasive Data Visualizations, Harvard Business Review Press, 2016.
- Evergreen SDH: Effective Data Visualization: The Right Chart for the Right Data, Sage Pubn, 2016.
- Foreman JW: Data Smart: Using Data Science to Transform Information into Insight, 1st edition, Wiley, 2013.
- Marr B: Big Data: Using SMART Big Data, Analytics and Metrics to Make Better Decisions and Improve Performance, 1st edition, Wiley, 2015.
- Nussbaumer Knafl C: Storytelling mit Daten. Die Grundlagen der effektiven Kommunikation und Visualisierung mit Daten, 1. Auflage, Vahlen, 2017.

Modulbeschreibung

Entrepreneurship

Modulnummer	Niveau Master	Studiengangsleitung Prof. Dr. Swati Chandna
ECTS (Benotung) 6 (benotet)	Dauer Semester	Häufigkeit 2 per Year
Sprache English	Workload Insgesamt: 150 Stunden - davon Kontaktzeit: 34 Stunden - davon Selbststudium: 116 Stunden - davon Praxis: 0 Stunden	Veranstaltungstyp Gruppenarbeit Projektarbeit

Teilnahmevoraussetzungen

Kursinformation

Modulprüfung

Prüfungsteile	Gewichtung (Bewertung)
Businessplan	70%
Präsentation	30%

Qualifikationsziele (Learning Outcomes) und Kompetenzen

Nach erfolgreichem Abschluss des Moduls können die Studierenden ...

Handlungskompetenz

- Develop a startup idea from scratch, validate it, and create a business model.
- Understand how to build and scale a company with a data-driven approach.
- Learn how to attract funding and investors through pitching and business planning.
- Explore customer acquisition and growth strategies tailored for AI/data-driven businesses.
- Identify legal, financial, and operational considerations in launching a company.
- Understand the AI startup ecosystem, including accelerators, funding rounds, and product-market fit.

Fachkompetenz „Wissen und Verstehen“

1. Fundamentals of Entrepreneurship

- How startups differ from traditional businesses
- Lean startup methodology & agile business development
- Business models: SaaS, B2B, B2C, marketplaces, and AI-as-a-Service

2. Market Research & Competitive Strategy

- Identifying market gaps and business opportunities
- Conducting competitive analysis using data-driven techniques
- Defining target customers & product-market fit

3. Financial Planning & Fundraising

- Basics of financial modeling & startup valuation
- Funding stages: bootstrapping, angel investors, venture capital
- Monetization strategies for AI & Data Science businesses

4. Building & Scaling a Tech Business

- Team formation & startup leadership for technical founders
- AI & data-driven decision-making for scaling operations
- Ethical considerations & responsible AI deployment in startups

Methodenkompetenz „Einsatz, Anwendung und Erzeugung von Wissen“

- Create a business model canvas for an AI or Data Science startup.
- Develop a financial projection model using real-world startup scenarios.
- Perform data-driven market research using publicly available datasets.
- Design a go-to-market strategy based on customer and competitor analysis.
- Practice pitching and investment strategies with real-world case studies

Selbstkompetenz „Wissenschaftliches Selbstverständnis und Professionalität“

- Critically assess the risks and opportunities of starting a business.
- Navigate ethical challenges and biases in AI-driven business models.
- Gain confidence in leading a team, making business decisions, and adapting to change.

Sozialkompetenz „Kommunikation und Kooperation“

- Work in interdisciplinary teams to develop a startup idea.
- Communicate business strategies to technical and non-technical stakeholders.
- Engage with mentors, investors, and startup founders through networking opportunities.

Constructive Alignment

1. Learning Outcomes Focus

- Entrepreneurship fundamentals for technical founders.
- Data-driven business decisions & financial strategy.
- Building, launching, and scaling a company as a Data Scientist.

2. Teaching & Learning Activities

- Lectures on business strategy, financial planning, and AI startup models.
- Workshops on creating business plans, pitch decks, and investor presentations.
- Case studies of successful (and failed) AI startups and their lessons.
- Group projects: developing a startup idea and pitching it to a panel.

Inhalt des Moduls

1. Introduction to Entrepreneurship & The Startup Ecosystem

- How startups work: venture capital, growth, and scaling
- Lean startup methodology and why businesses fail

2. Business Model Development

- Business Model Canvas & Value Proposition Canvas
- Monetization models for tech & AI startups

3. Market Research & Competitive Strategy

- Data-driven market research using public datasets
- Customer segmentation & acquisition strategies

4. Financial Planning & Investment

- Startup funding rounds & investor expectations
- Building a simple financial model for a startup

5. Building & Scaling an AI/Data-Driven Business

- Hiring a team & leading a startup as a technical founder
- AI ethics & responsible AI deployment in real-world applications

6. Go-to-Market Strategy & Growth Hacking

- Digital marketing, sales funnels, and customer retention strategies
- AI-powered automation for scaling operations

7. The Startup Pitch: How to Raise Funds

- Writing a compelling pitch deck
- How to negotiate with investors & secure funding

8. Case Studies & Real-World Applications

- Elon Musk's approach to scaling Tesla, OpenAI, and SpaceX
- AI startup failures & lessons from Theranos, WeWork, and others
- How AI entrepreneurs are disrupting industries today

Literatur

- Ries, E. (2011). The Lean Startup: How Today's Entrepreneurs Use Continuous Innovation. Crown Business.
- Christensen, C. (2016). The Innovator's Dilemma. Harvard Business Review Press.
- Blank, S. (2020). The Startup Owner's Manual: The Step-by-Step Guide for Building a Great Company. Wiley.
- Thiel, P. (2014). Zero to One: Notes on Startups, or How to Build the Future. Currency.
- Maurya, A. (2016). Scaling Lean: Mastering the Key Metrics for Startup Growth. Portfolio.



Semester 3: General or Interdisciplinary Track

Campus Specific



General Track

Heidelberg

Modulbeschreibung

AI Vision

Modulnummer	Niveau Master	Studiengangsleitung Prof. Dr. Swati Chandna
ECTS (Benotung) 6 (benotet)	Dauer Semester	Häufigkeit 2 per Year
Sprache English	Workload Insgesamt: 150 Stunden - davon Kontaktzeit: 64 Stunden - davon Selbststudium: 86 Stunden - davon Praxis: 0 Stunden	Veranstaltungstyp Lehrvortrag Tutorium Übung

Teilnahmevoraussetzungen

Kursinformation

Modulprüfung

Prüfungsteile	Gewichtung (Bewertung)
Praktische Arbeit	100%

Qualifikationsziele (Learning Outcomes) und Kompetenzen

Nach erfolgreichem Abschluss des Moduls können die Studierenden ...

Handlungskompetenz

This module provides a comprehensive introduction to the field of computer vision, covering fundamental concepts, core techniques, and cutting-edge advancements in AI-powered visual understanding. Students will gain both theoretical knowledge and practical experience in designing, implementing, and evaluating computer vision systems for a variety of applications, with a strong emphasis on deep learning methods.

Fachkompetenz „Wissen und Verstehen“

Students completing this course will possess the following knowledge:

- Explain the fundamentals of computer vision, image formation, and basic image processing.
- Understand deep learning for image classification, including CNN architectures and training.
- Describe object detection/recognition techniques: region-based CNNs, SSDs, and YOLO.
- Explain image segmentation: semantic, instance, and panoptic segmentation.
- Understand 3D computer vision principles: depth estimation, stereo vision, point clouds.
- Describe methods for face detection and recognition (traditional and deep learning).
- Understand generative models (GANs, VAEs) for image generation and manipulation.
- Explain Vision Transformers and Multimodal Learning (Vision-Language Models).

Methodenkompetenz „Einsatz, Anwendung und Erzeugung von Wissen“

Through practical exercises and a culminating project, students will be able to:

- Implement basic image processing using libraries.
- Train and evaluate deep learning models for image classification.
- Implement and apply object detection/recognition algorithms.
- Utilize image segmentation techniques.
- Apply 3D computer vision techniques for depth estimation.
- Develop and evaluate face detection/recognition systems.
- Implement and experiment with generative models for image synthesis.
- Critically evaluate, compare, implement and fine-tune Vision Transformers and Multimodal models.

Selbstkompetenz „Wissenschaftliches Selbstverständnis und Professionalität“

This module will encourage students to:

- Independently research and stay up-to-date with the latest advancements in computer vision.
- Critically assess the ethical implications of computer vision technologies, such as bias in datasets and potential for misuse.
- Manage time and resources effectively when working on computer vision projects.
- Develop a problem-solving mindset and a systematic approach to addressing challenges in computer vision.
- Take initiative in exploring new techniques and applications beyond the core course material.

Sozialkompetenz „Kommunikation und Kooperation“

This module facilitates developing social skills:

- Communicate technical concepts and results clearly and concisely, both orally and in writing.
- Present project findings and defend design choices in a professional manner.
- Engage in constructive discussions and provide/receive feedback on technical topics.
- Seek input from instructors and peers to improve.

Constructive Alignment

The learning outcomes, course content, and assessment methods in this module are designed for a cohesive and

effective learning experience. Theoretical lectures are combined with hands-on practical sessions. The primary assessment is Practical Work, requiring students to demonstrate mastery of computer vision concepts and techniques by improving upon existing research:

- Research Paper Selection: Select a recent, peer-reviewed research paper focused on a computer vision task, utilizing a publicly available dataset, and submit it for instructor approval.
- Dataset Understanding and Preparation: Obtain, analyze, and prepare the chosen dataset, documenting their understanding and any preprocessing steps.
- Baseline Model Replication: Implement the method/model from the selected paper to establish a baseline for comparison, aiming to replicate the published results.
- Model/Method Improvement: Develop and implement an improved model/method, striving to demonstrably outperform the baseline through various techniques.
- Evaluation: Evaluate both baseline and improved models/methods using appropriate metrics and statistical significance testing on a held-out test set.
- Documentation: Prepare a detailed report documenting the entire project, including paper summary, dataset analysis, model descriptions, results, and critical discussion.
- Code Submission: Submit well-documented, clean, and reproducible code for both the baseline and improved implementations.

Inhalt des Moduls

1. Introduction to AI Vision

- History of computer vision
- Traditional vs. AI-driven approaches

2. Image Processing Fundamentals

- Image formats and color models
- Filtering, edge detection, contour analysis
- Histogram equalization, thresholding, morphological operations

3. Convolutional Neural Networks (CNNs)

- Architecture and convolution principles
- Pooling, activation, normalization
- Architectures: LeNet, AlexNet, VGG, ResNet, EfficientNet

4. Image Classification

- Transfer learning and fine-tuning
- Pre-trained models (e.g., ImageNet, EfficientNet)
- Evaluation metrics: accuracy, confusion matrix

5. Object Detection

- Sliding window and region proposals (R-CNN)
- Fast/Faster R-CNN, SSD, YOLO (You Only Look Once)
- Real-time detection and annotation

6. Semantic and Instance Segmentation

- Difference between segmentation types
- U-Net, Mask R-CNN, DeepLab
- Medical imaging and satellite data examples

7. Generative Vision Models

- Autoencoders, Variational Autoencoders (VAEs)
- Generative Adversarial Networks (GANs)

8. Vision Transformers (ViT)

- Self-attention in vision tasks
- ViT vs. CNNs

9. 3D Vision and Depth Estimation

- Stereo vision, point clouds, SLAM
- LiDAR and depth sensors

10. Video Analysis and Action Recognition

- Frame-level vs. sequence-level analysis
- 3D CNNs, LSTM/GRU for vision

11. Multimodal Vision (Vision + Language)

- Image captioning (CNN + RNN, Transformer-based models)
- Visual Question Answering (VQA)
- CLIP, BLIP, Flamingo, and multi-modal LLMs

12. Explainability, Ethics, and Bias in AI Vision

- Explainable AI for vision
- Dataset bias, adversarial attacks
- Ethical deployment and societal impact

Literatur

- Szeliski, R. (2022). Computer Vision: Algorithms and Applications (2nd ed.). Springer.
- Goodfellow, I., Bengio, Y., & Courville, A. (2016). Deep Learning. MIT Press.
- Prince, S. J. D. (2012). Computer Vision: Models, Learning, and Inference. Cambridge University Press.
- Forsyth, D. A., & Ponce, J. (2012). Computer Vision: A Modern Approach (2nd ed.). Pearson.
- Gonzalez, R. C., & Woods, R. E. (2017). Digital Image Processing (4th ed.). Pearson.
- Hartley, R., & Zisserman, A. (2003). Multiple View Geometry in Computer Vision (2nd ed.). Cambridge University Press.
- Shapiro, L. G., & Stockman, G. C. (2001). Computer Vision. Prentice Hall.
- Davies, E. R. (2017). Computer Vision: Principles, Algorithms, Applications, Learning (5th ed.). Academic Press.
- Burger, W., & Burge, M. J. (2007). Principles of digital image processing: fundamental techniques. Springer Science & Business Media.
- Jain, R., Kasturi, R., & Schunck, B. G. (1995). Machine vision. McGraw-Hill, Inc.
- Bradski, G., & Kaehler, A. (2008). Learning OpenCV: Computer vision with the OpenCV library. " O'Reilly Media, Inc."



Creative AI and Media Analytics

Heidelberg

Modulbeschreibung

Gaming Analytics

Modulnummer	Niveau Master	Studiengangsleitung Prof. Dr. Swati Chandna
ECTS (Benotung) 6 (benotet)	Dauer Semester	Häufigkeit 2 per Year
Sprache English	Workload Insgesamt: 150 Stunden - davon Kontaktzeit: 34 Stunden - davon Selbststudium: 116 Stunden - davon Praxis: 0 Stunden	Veranstaltungstyp Austausch Gruppenarbeit Präsentation

Teilnahmevoraussetzungen

Kursinformation

Modulprüfung

Prüfungsteile	Gewichtung (Bewertung)
Projektarbeit	100%

Qualifikationsziele (Learning Outcomes) und Kompetenzen

Nach erfolgreichem Abschluss des Moduls können die Studierenden ...

Handlungskompetenz

This module introduces students to the field of Gaming Analytics, which focuses on extracting actionable insights from data generated in digital games. Students will explore how analytics supports game design, player engagement, monetization, and live operations, applying statistical, machine learning, and behavioral modeling techniques to real-world gaming datasets. The course balances theoretical foundations, data modeling, tool use, and ethical considerations in the evolving games industry.

Fachkompetenz „Wissen und Verstehen“

- Understand the role of data across the game development lifecycle (pre-launch, live ops, postmortem)
- Model player behavior using telemetry data, session metrics, and progression logs
- Design and evaluate key performance indicators (KPIs) such as retention, churn, and LTV
- Analyze game economy balance, monetization funnels, and player segmentation

Methodenkompetenz „Einsatz, Anwendung und Erzeugung von Wissen“

- Apply data wrangling, clustering, regression, and survival analysis to gameplay data
- Use tools like Python (pandas, seaborn, scikit-learn), SQL, and Tableau/Power BI
- Design AB tests and telemetry instrumentation for new game features
- Build dashboards to monitor player activity, engagement, and economy health

Selbstkompetenz „Wissenschaftliches Selbstverständnis und Professionalität“

- Reflect on ethical implications of behavioral targeting and monetization strategies
- Develop critical thinking around game fairness, dark patterns, and player well-being
- Take initiative in proposing data-informed design improvements or retention strategies

Sozialkompetenz „Kommunikation und Kooperation“

- Collaborate in multidisciplinary teams involving data scientists, designers, and product managers
- Communicate analytical findings to both technical and creative stakeholders
- Align analytics with player empathy and design intent

Constructive Alignment

- Lectures and lab sessions build theoretical foundations and applied tool proficiency (Specialist + Methodological)
- Case study reviews and design critiques promote design thinking and empathy with game players (Social + Personal)
- Hands-on analytics projects allow students to apply modeling, clustering, and dashboarding tools to player data
- Peer collaboration and presentation activities encourage team-based decision-making and stakeholder communication

Inhalt des Moduls

1. Introduction to Gaming Analytics

- Overview of the games industry and game types (casual, AAA, mobile, live-service)
- Analytics use cases across departments: design, marketing, monetization, support
- Data sources in games: events, telemetry, player logs, social interactions

2. Game Metrics and Player KPIs

- Core metrics: DAU, MAU, session length, ARPDAU, conversion rate
- Retention curves and cohort analysis
- Funnel analysis and drop-off points
- LTV (lifetime value) modeling and payback periods

3. Player Behavior Modeling

- Session segmentation, heatmaps, and behavioral states
- Player journey mapping and churn prediction
- Clustering players by progression or spend patterns
- Engagement typologies and achievement paths

4. Game Economy and Balance Analytics

- In-game currencies, sinks, and faucets
- Virtual item valuation and scarcity modeling
- Economy inflation and deflation trends
- Simulation of balance adjustments using past data

5. A/B Testing and Feature Impact

- Experimental design in games
- Measuring impact of tutorials, mechanics, pricing, or cosmetics
- Sample sizing and statistical power
- Interpreting significance in behavioral data

6. Monetization and Ethical Analytics

- Monetization models: ads, IAP, subscriptions, battle passes
- Behavioral economics in games: nudges, compulsion loops
- Ethics: pay-to-win vs fair monetization, loot boxes, youth protections
- Regulatory trends: GDPR, PEGI, ESRB considerations

7. Real-Time Game Telemetry and Dashboards

- Collecting, storing, and streaming player event data
- Designing dashboards for live ops, alerts, and performance tracking
- Reporting strategies for game producers and community managers

8. AI and Predictive Analytics in Games

- Churn prediction using supervised ML
- Recommender systems for in-game content
- Adaptive difficulty tuning with reinforcement learning (intro)
- Anomaly detection for cheating, bugs, or revenue drops

Literatur

- Seif El-Nasr, M., Drachen, A., & Canossa, A. (2016). Game Analytics: Maximizing the Value of Player Data
- Fields, T., & Cotton, B. (2014). Social Game Design: Monetization Methods and Mechanics
- Rouse, R. (2022). Game Design: Theory and Practice (for linking gameplay to analytics)
- Kim, J. (2022). Behavioral Game Analytics: Player Modeling and Retention



Supply Chain and Logistics Analytics

Hamburg

Modulbeschreibung

Risk and Sustainability Analytics

Modulnummer	Niveau Master	Studiengangsleitung Prof. Dr. Swati Chandna
ECTS (Benotung) 6 (benotet)	Dauer Semester	Häufigkeit 2 per Year
Sprache English	Workload Insgesamt: 150 Stunden - davon Kontaktzeit: 34 Stunden - davon Selbststudium: 116 Stunden - davon Praxis: 0 Stunden	Veranstaltungstyp Fallarbeit Gruppenarbeit Lehrvortrag

Teilnahmevoraussetzungen

Kursinformation

Modulprüfung

Prüfungsteile	Gewichtung (Bewertung)
Projektarbeit	70%
Präsentation	30%

Qualifikationsziele (Learning Outcomes) und Kompetenzen

Nach erfolgreichem Abschluss des Moduls können die Studierenden ...

Handlungskompetenz

This module enables students to apply data science and analytical techniques to assess, monitor, and manage environmental, social, governance (ESG) and operational risks in business and policy contexts. Students will develop critical competencies in scenario modeling, sustainability KPIs, climate risk assessment, and resilience forecasting, preparing them to support data-informed decisions in line with global sustainability standards and risk governance frameworks.

Fachkompetenz „Wissen und Verstehen“

- Understand core principles of sustainability (ESG, SDGs, triple bottom line) and enterprise risk management
- Apply data analytics to quantify climate, social, financial, and supply chain risks
- Evaluate sustainability performance through indicators (carbon footprint, water stress, GRI metrics)
- Build risk matrices, materiality maps, and sustainability heatmaps

Methodenkompetenz „Einsatz, Anwendung und Erzeugung von Wissen“

- Apply forecasting (time series), simulation (Monte Carlo), and classification models to risk scenarios
- Use dashboards and geospatial visualizations for sustainability reporting
- Work with ESG frameworks and scenario planning (TCFD, EU Taxonomy, SASB, etc.)
- Analyze climate-related data (e.g., emissions, disaster frequency, resource consumption)

Selbstkompetenz „Wissenschaftliches Selbstverständnis und Professionalität“

- Reflect on personal and societal responsibilities in managing systemic risks
- Understand the role of data bias and uncertainty in decision-making
- Develop sustainable thinking and resilience mindsets in professional contexts

Sozialkompetenz „Kommunikation und Kooperation“

- Collaborate in interdisciplinary teams on real-world sustainability and risk problems
- Communicate risks and sustainability findings to stakeholders (investors, regulators, communities)
- Participate in ethical and political discussions around carbon trading, greenwashing, and equity

Constructive Alignment

- Lectures and case-based exercises provide foundations in risk and sustainability theory (Specialist)
- Dashboards, simulations, and forecasting labs build data modeling and communication skills (Methodological)
- Capstone project and roleplay pitch develop team collaboration and stakeholder engagement (Social)
- Ethics modules and self-reflection support responsible leadership and awareness of systemic impacts (Personal)

Inhalt des Moduls

1. Introduction to Risk and Sustainability Analytics

- Definitions: risk vs uncertainty, sustainability dimensions (social, environmental, economic)
- Introduction to global standards: SDGs, TCFD, GRI, SASB, EU Green Deal
- Data types: structured vs unstructured, climate vs financial data sources

2. Risk Identification and Quantification

- Risk typologies: operational, environmental, financial, reputational
- Risk registers, bowtie diagrams, and impact-probability matrices
- Historical risk data analysis and trend extraction

3. Scenario Analysis and Forecasting

- Climate scenario modeling (RCPs, SSPs)
- Monte Carlo simulation for financial and climate risk
- Time series models for disaster frequency, emissions, or demand forecasting
- Forecast error evaluation and uncertainty quantification

4. Sustainability Dashboards and Decision Support

- Designing dashboards for ESG reporting using Power BI/Tableau
- KPI visualization and materiality matrix construction
- Integration with sustainability reports (GRI, integrated reporting)
- Geo-mapping risks (flood zones, deforestation, pollution hotspots)

Literatur

- Treitl, S. (2021). AI in Logistics: Machine Learning and Optimization. Springer.
- Laporte, G. (2019). Transportation Logistics and Network Flow Models. Wiley.
- Silver, E. A., Pyke, D. F., & Peterson, R. (2016). Inventory Management and Production Planning. Wiley.
- Russell, S. & Norvig, P. (2020). Artificial Intelligence: A Modern Approach. Pearson.
- Goodfellow, I., Bengio, Y., & Courville, A. (2016). Deep Learning. MIT Press.
- Bertsimas, D. & Tsitsiklis, J. (1997). Introduction to Linear Optimization. Athena Scientific.
- Elkington, J. (1999). Cannibals with Forks: The Triple Bottom Line of 21st Century Business
- WEF & PwC (2021). Measuring Stakeholder Capitalism: ESG Metrics for Sustainable Value Creation
- Alexander, C. & Sheedy, E. (2005). The Professional Risk Managers' Handbook
- IPCC Reports, TCFD Recommendations, GRI Standards



Business Analytics

Fuerth

Modulbeschreibung

Strategic Communication and Leadership

Modulnummer	Niveau Master	Studiengangsleitung Prof. Dr. Clemens Werkmeister
ECTS (Benotung) 6 (benotet)	Dauer Semester	Häufigkeit 2 per Year
Sprache English	Workload Insgesamt: 150 Stunden - davon Kontaktzeit: 34 Stunden - davon Selbststudium: 116 Stunden - davon Praxis: 0 Stunden	Veranstaltungstyp Gruppenarbeit Interaktives Seminar

Teilnahmevoraussetzungen

Kursinformation

Modulprüfung

Prüfungsteile	Gewichtung (Bewertung)
Präsentation	100%

Qualifikationsziele (Learning Outcomes) und Kompetenzen

Nach erfolgreichem Abschluss des Moduls können die Studierenden ...

Handlungskompetenz

Students will be able to lead teams and projects effectively by applying principles of strategic communication and leadership. They will develop the competence to align communication strategies with organizational goals and to influence stakeholders in data-driven business environments.

Fachkompetenz „Wissen und Verstehen“

Students understand the key concepts of leadership theory, strategic communication, and change management. They can analyze and evaluate communication flows and leadership behavior in organizational and intercultural contexts.

Methodenkompetenz „Einsatz, Anwendung und Erzeugung von Wissen“

Students acquire the ability to apply communication models, leadership diagnostics, and stakeholder analysis tools. They learn to design communication strategies tailored to various audiences, including data storytelling and persuasive presentation techniques.

Selbstkompetenz „Wissenschaftliches Selbstverständnis und Professionalität“

Students reflect on their personal leadership style and communication habits, developing greater self-awareness and emotional intelligence. They cultivate resilience and adaptability in complex decision-making environments.

Sozialkompetenz „Kommunikation und Kooperation“

Students improve their interpersonal skills in team settings, including conflict resolution, active listening, and constructive feedback. They learn to lead diverse teams with empathy and to engage stakeholders through inclusive and ethical communication.

Constructive Alignment

The module integrates leadership theory with applied communication practice. Students engage in simulations, role plays, and case studies that mirror real-world strategic communication challenges. Assessment is based on practical assignments, including leadership self-reflection and the development of a strategic communication plan.

Inhalt des Moduls

- Foundations of leadership theory and leadership styles
- Principles of strategic and organizational communication
- Communication planning and stakeholder mapping
- Data storytelling and communicating insights effectively
- Managing communication in change and crisis situations
- Leadership in diverse and virtual teams
- Influence, negotiation, and decision-making techniques
- Intercultural communication and global leadership
- Ethical leadership and communication responsibility
- Self-leadership and personal development planning

Literatur

- Northouse, P. G. (2021). Leadership: Theory and Practice (9th ed.). Sage Publications.
- Veltsos, J.R. (2024). Managerial Communication (9th ed.). Sage Publications.
- Goleman, D. (2006). Emotional Intelligence: Why It Can Matter More Than IQ. Bantam Books.
- Duarte, N. (2010). Resonate: Present Visual Stories that Transform Audiences. Wiley.
- Hoffman, H. (2025). Leadership Strategy & Tactics Made Simple.
- Iansiti, M.; Lakhani, K.M. (2020): Competing in the Age of AI: Strategy and Leadership. Harvard Business Review Press.



AI-Driven Bioinformatics & Life Sciences (Munich)

Munich

Modulbeschreibung

Biomedical image analysis

Modulnummer	Niveau Master	Studiengangsleitung Prof. Dr. Swati Chandna
ECTS (Benotung) 6 (benotet)	Dauer Semester	Häufigkeit 2 per Year
Sprache English	Workload Insgesamt: 150 Stunden - davon Kontaktzeit: 34 Stunden - davon Selbststudium: 116 Stunden - davon Praxis: 0 Stunden	Veranstaltungstyp Diskussion Gruppenarbeit

Teilnahmevoraussetzungen

Kursinformation

Modulprüfung

Prüfungsteile	Gewichtung (Bewertung)
Praktische Arbeit	100%

Qualifikationsziele (Learning Outcomes) und Kompetenzen

Nach erfolgreichem Abschluss des Moduls können die Studierenden ...

Handlungskompetenz

This module introduces students to the field of biomedical image analysis, focusing on how computer vision, machine learning, and deep learning techniques are applied to extract meaningful information from medical and biological imaging data. Students will work with modalities such as MRI, CT, histopathology, and microscopy, using algorithms for preprocessing, segmentation, classification, and anomaly detection. The course emphasizes both technical skills and ethical considerations in clinical and biomedical research contexts.

Fachkompetenz „Wissen und Verstehen“

- Understand medical image modalities (MRI, CT, PET, histology) and their diagnostic relevance
- Apply image preprocessing, segmentation, and enhancement techniques to biomedical data
- Evaluate classification, detection, and annotation pipelines in a biomedical context

Assess image quality, variability, and annotation reliability in medical datasets

Methodenkompetenz „Einsatz, Anwendung und Erzeugung von Wissen“

- Use tools like OpenCV, scikit-image, SimpleITK, and PyTorch/TensorFlow for image analysis
- Implement pipelines for denoising, filtering, morphological operations, and registration
- Train and evaluate deep learning models (e.g., CNNs, U-Net, ResNet) on biomedical images
- Apply evaluation metrics (Dice coefficient, IoU, sensitivity/specificity)

Selbstkompetenz „Wissenschaftliches Selbstverständnis und Professionalität“

- Reflect on the ethical and legal implications of medical imaging AI (e.g., bias, privacy, fairness)
- Demonstrate accountability in handling sensitive patient and clinical data
- Critically evaluate limitations of algorithms in high-stakes biomedical decision-making

Sozialkompetenz „Kommunikation und Kooperation“

- Collaborate in interdisciplinary teams with biologists, radiologists, and data scientists
- Communicate model results to clinicians and biomedical stakeholders
- Address issues of clinical trust, explainability, and medical safety

Constructive Alignment

- Lectures and tool demos introduce theory, modalities, and workflows (Specialist)
- Labs and coding assignments build applied skills in biomedical image preprocessing and modeling (Methodological)
- Group projects and presentations encourage interdisciplinary collaboration and clinical communication (Social)
- Ethics discussions and reflections foster awareness of fairness, safety, and data privacy (Personal)

Inhalt des Moduls

1. Introduction to Biomedical Imaging
 - Overview of imaging modalities (MRI, CT, PET, Ultrasound, X-ray, Histology)
 - Biomedical data sources: NIH, TCIA, Allen Brain Atlas, CAMELYON
 - DICOM format and metadata interpretation
2. Image Preprocessing and Enhancement
 - Image filtering, normalization, histogram equalization
 - Noise reduction (Gaussian, median filtering)
 - Contrast enhancement and standardization
 - Image registration and alignment techniques
3. Image Segmentation Techniques
 - Thresholding and region growing
 - Watershed segmentation
 - Machine learning-based segmentation
 - Deep learning with U-Net and variants
4. Feature Extraction and Classification
 - Texture, shape, intensity-based features
 - Feature selection and dimensionality reduction (PCA, t-SNE)
 - Classification using traditional ML (SVM, Random Forest, k-NN)
 - CNN-based classification in histopathology and radiology
5. Deep Learning for Biomedical Imaging
 - CNNs for image classification and detection
 - U-Net, V-Net, and 3D CNNs for medical image segmentation
 - Transfer learning and data augmentation
 - Performance evaluation and medical model calibration
6. Multi-Modal and 3D Imaging
 - 3D imaging data handling and visualization
 - Multimodal fusion (e.g., MRI + genomics, PET + CT)
 - Volumetric analysis and reconstruction
7. Annotation, Datasets, and Weak Supervision
 - Ground truth generation and inter-observer variability
 - Semi-supervised and weakly supervised learning
 - Dealing with label noise and small datasets
8. Explainability and Model Validation
 - Saliency maps, Grad-CAM, SHAP for imaging models
 - Cross-validation, bootstrapping, and external validation
 - Model uncertainty and clinical decision support

Literatur

- Gonzalez & Woods. Digital Image Processing (Medical extensions)
- Zhou, S. K., Greenspan, H., & Shen, D. (2019). Deep Learning for Medical Image Analysis

-Shilo, S., & Benjamini, Y. (2022). Statistical Challenges in Modern Medical Imaging



Semester 4: Mandatory Modules

Heidelberg, Hamburg, München, Fürth

Modulbeschreibung

Master Thesis Seminar

Modulnummer	Niveau Master	Studiengangsleitung Prof. Dr. Swati Chandna
ECTS (Benotung) 6 (benotet)	Dauer Semester	Häufigkeit
Sprache English	Workload Insgesamt: 0 Stunden - davon Kontaktzeit: 0 Stunden - davon Selbststudium: 0 Stunden - davon Praxis: 0 Stunden	Veranstaltungstyp Diskussion Gruppenarbeit Literaturrecherche

Teilnahmevoraussetzungen

Kursinformation

Modulprüfung

Prüfungsteile	Gewichtung (Bewertung)
Exposé	100%

Qualifikationsziele (Learning Outcomes) und Kompetenzen

Nach erfolgreichem Abschluss des Moduls können die Studierenden ...

Handlungskompetenz

This seminar supports students in preparing, structuring, and communicating their master's thesis research. It emphasizes academic writing, scientific argumentation, methodological alignment, and research ethics. Students will develop the competence to critically reflect on their thesis progress, engage with feedback, and present their work clearly to both academic and applied audiences.

Fachkompetenz „Wissen und Verstehen“

- Develop and refine a research question aligned with data science and AI domains
- Structure a rigorous methodology appropriate for the thesis topic
- Conduct critical literature reviews and synthesize findings in context

Methodenkompetenz „Einsatz, Anwendung und Erzeugung von Wissen“

- Apply academic writing conventions and citation standards
- Prepare reproducible research documentation (e.g., notebooks, Git repos, data statements)
- Use thesis templates and version control tools (e.g., Overleaf, LaTeX, GitHub)

Selbstkompetenz „Wissenschaftliches Selbstverständnis und Professionalität“

- Manage independent research timelines and complexity
- Demonstrate critical self-assessment and academic integrity
- Respond constructively to feedback and iterate based on reflection

Sozialkompetenz „Kommunikation und Kooperation“

- Participate in scholarly discussion and peer feedback
- Present intermediate and final research to diverse academic audiences
- Collaborate with advisors and peers to improve research quality

Constructive Alignment

- Seminar discussions and interim presentations build research fluency (Specialist)
- Thesis proposal and planning strengthen methodological awareness and execution (Methodological)
- Peer review and group sessions foster academic dialogue (Social)
- Ethical discussions and reflection journals reinforce independence and accountability (Personal)

Inhalt des Moduls

1. Research Framing and Topic Finalization

- Structuring the research question, objectives, and scope
- Aligning with scientific and societal relevance

2. Literature Review and Theoretical Grounding

- Conducting and documenting systematic/narrative literature reviews
- Identifying gaps and positioning the thesis

3. Methodology, Data, and Ethics

- Matching research methods to questions (quantitative, qualitative, mixed)
- Data access, privacy, reproducibility, and risk mitigation
- Research ethics and approval processes (if applicable)

4. Scientific Writing and Reporting

- Structuring the thesis document (abstract to conclusion)
- Referencing systems (APA, IEEE) and citation managers
- Avoiding plagiarism and self-plagiarism

5. Presenting Research

- Designing visuals: charts, diagrams, and dashboards
- Presentation rehearsal with peer and faculty feedback
- Defense simulation and Q&A strategies

Literatur

- Booth et al. – The Craft of Research
- Silvia – How to Write a Lot
- UMUC – Master Thesis Template (LaTeX or Word)
- Zotero / Mendeley for citations
- GitHub, Overleaf for collaborative writing and documentation

Modulbeschreibung

Master Thesis

Modulnummer	Niveau Master	Studiengangsleitung Prof. Dr. Swati Chandna
ECTS (Benotung) 24 (benotet)	Dauer Semester	Häufigkeit
Sprache English	Workload Insgesamt: 0 Stunden - davon Kontaktzeit: 0 Stunden - davon Selbststudium: 0 Stunden - davon Praxis: 0 Stunden	Veranstaltungstyp Einzelarbeit Kolloquium Präsentation

Teilnahmevoraussetzungen

Kursinformation

Modulprüfung

Prüfungsteile	Gewichtung (Bewertung)
Thesis	75%
Kolloquium	25%

Qualifikationsziele (Learning Outcomes) und Kompetenzen

Nach erfolgreichem Abschluss des Moduls können die Studierenden ...

Handlungskompetenz

The Master Thesis module allows students to independently conduct an applied or theoretical research project under academic supervision. It provides an opportunity to demonstrate advanced domain knowledge, methodological competence, and the ability to generate novel insights in data science and AI or its application areas. The thesis must meet academic standards of clarity, rigor, and ethical responsibility.

Fachkompetenz „Wissen und Verstehen“

- Apply core knowledge from data science, AI, or a specific application domain to a defined research problem
- Formulate a precise and feasible research question and objectives
- Demonstrate in-depth understanding of relevant theories and scientific discourse

Methodenkompetenz „Einsatz, Anwendung und Erzeugung von Wissen“

- Design and execute a reproducible research process, including data collection, analysis, and interpretation
- Select and apply suitable analytical, statistical, computational, or modeling techniques
- Document code, data, and workflow using professional tools (e.g., GitHub, Jupyter, LaTeX/Overleaf)

Selbstkompetenz „Wissenschaftliches Selbstverständnis und Professionalität“

- Demonstrate research autonomy, responsibility, and time management
- Uphold ethical standards, academic integrity, and legal compliance (e.g., GDPR)
- Reflect critically on one's research process, limitations, and societal impact

Sozialkompetenz „Kommunikation und Kooperation“

- Communicate findings clearly to academic and applied audiences
- Engage in constructive dialogue with supervisors, peers, or industry partners
- Position one's work within the broader scientific and societal context

Constructive Alignment

- Independent research and writing build expert-level knowledge and domain immersion (Specialist)
- Application of research methodology and computational tools demonstrate analytical rigor (Methodological)
- Presentation and dialogue with evaluators support scientific communication and positioning (Social)
- Research planning, iteration, and ethics cultivate self-awareness and scholarly discipline (Personal)

Inhalt des Moduls

Required Deliverables

1. Master Thesis Document

- Typically 60–80 pages (excluding appendices)
- Includes abstract, introduction, literature review, methodology, results, discussion, conclusion
- Must follow formatting and citation standards (APA, IEEE, etc.)

2.Oral Defense (Colloquium)

- Presentation (15 minutes) + Q&A
- Evaluated based on clarity, scientific argumentation, and ability to respond to critique

3. Optional Supplementary Materials

- Code repositories, data samples, notebooks
- Ethics documentation (e.g., consent forms, data protection forms if applicable)

Literatur

- Booth et al. – The Craft of Research
- Rowena Murray – How to Write a Thesis
- Creswell – Research Design: Qualitative, Quantitative, and Mixed Methods
- GitHub/GitLab – for version control and open science
- Jupyter/Overleaf – for documenting code and research writing